

The Standard Model Carrier Exhaustion Closure

Primitive Closure, Skin Openness, BSM Audit, and No Same-Scope Primitive Escape

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Abstract

This paper completes the Standard Model carrier-exhaustion sequence by asking whether any same-scope Standard Model-associated attribute retains primitive standing outside the certified carrier package once the sector closures have been received. The method is an AASC primitive-status exhaustion proof: it distinguishes primitive carrier standing from derivative skins, routes, projections, numerical statuses, evidence records, BSM surfaces, and target updates under the kernel conditions required for non-degenerate physical claimhood.

The proof receives the master classifier, the gauge-sector closure, the matter-sector closure, the charge/coupling/renormalization closure, the flavor/mixing/generation closure, the open-generation cardinality endpoint, the Higgs bridge/fixation endpoint, the kernel/admissibility source, and the standalone Standard Model structural spine. The kernel is not an interpretive overlay placed on physics. It is the minimal non-degeneracy condition under which physical claims retain standing, reference, sameness, comparison class, prediction, and falsification.

With

$$C_{SM}^{\text{prim}} = C_{\text{gauge}} \cup C_{\text{matter}} \cup C_{CCR}^{\text{prim}} \cup C_{\text{flavor}}^{\text{prim}} \cup \{H_{br/fix}\},$$

the main result is the empty-survivor theorem

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset.$$

The result is not empirical finality, not a no-new-physics thesis, not a derivation of all numerical Standard Model parameters, and not an empirical family-count selector. It proves instead that, relative to the fixed same-scope Standard Model carrier arena with generation governance instantiated by the open-generation endpoint, no attribute outside the certified primitive carrier package closes as a residual primitive.

The theorem is falsifiable in its proof class. A valid same-scope primitive outside C_{SM}^{prim} , neither derivative nor registered nor generation-governed nor target-updating, would defeat SM-6

by breaking a received closure input, constructor adequacy, role-necessity exhaustion, target classification, or an SM-6 bridge lemma. Under the fixed dependency package and constructor-presented arena, no such hostile certificate closes. The Standard Model is therefore primitively closed at the same-scope carrier level while remaining open to new measurements, refined fits, EFT displays, anomalies, BSM audit cases, projection languages, and target-updating physics.

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1 Introduction: the closure question

1.1 The problem

The Standard Model admits many kinds of associated descriptions: gauge-field symbols, quark and lepton labels, color and weak charges, coupling values, running curves, mass parameters, CKM and PMNS entries, oscillation routes, Higgs scalar notation, vacuum branches, EFT coefficients, detector records, anomaly reports, and BSM model surfaces. The question of this paper is not whether these descriptions are scientifically meaningful. They are. The question is whether any such description retains *primitive carrier standing* inside the fixed same-scope Standard Model arena once the local carrier sectors have been closed.

The closure problem is therefore precise. Once gauge standing, matter standing, charge/coupling/renormalization standing, flavor transition standing, generation governance, and Higgs bridge/fixation standing have received their sector endpoints, is there a residual Standard Model-associated primitive outside the certified carrier package? SM-6 answers no, under the received dependency package and the global bridge lemmas proved here.

The target is not a physics encyclopedia. It is a primitive-status exhaustion proof over the non-degeneracy conditions required for same-scope Standard Model claims to retain standing, reference, comparison, prediction, and falsification. Its object is the primitive-status ledger of the Standard Model carrier domain. It separates the primitive carriers required for same-scope standing from the downstream skins, routes, projections, coordinates, measurements, and update branches that are licensed by, measured against, or compared through those carriers.

1.2 Primitive closure is not empirical finality

The endpoint asserted here must not be weakened into mere classifier convenience, but it must also not be inflated into empirical finality. The paper does not claim that the Standard Model is the final physics of nature. It does not claim that BSM physics is impossible. It does not claim that all Standard Model parameters are numerically derived. It does not infer generation cardinality from empirical family labels, matrix dimension, fitted textures, or fourth-generation non-observation; it receives the separate open-generation endpoint $\text{ClosedGenCert}(n) \iff n = 3$. It does not claim that EFT language, anomaly reports, detector data, renormalization schemes, or model-building surfaces are useless.

The claim is narrower and stronger. Within D_{SM}^3 , no Standard Model-associated attribute outside C_{SM}^{prim} survives as an unlicensed primitive. Familiar objects may remain indispensable as representatives, coordinates, routes, measurements, or projections; they do not thereby become primitive carriers. Future physics may alter measurements, add regimes, introduce new fields, or update the target. Such events may be scientifically decisive. They become same-scope primitive insertions into

the closed Standard Model carrier domain only by supplying a valid same-scope hostile certificate.

1.3 Primitive closure, skin openness

The controlling line is:

primitive closure, skin openness.

Primitive closure means

$$\text{Survprim}_D(\text{Attr}(D) \setminus C^{\text{prim}}) = 0,$$

for the specified domain and carrier package. Skin openness means that a theory can remain open at the level of values, representatives, measurements, computational routes, projection displays, effective descriptions, anomaly surfaces, and target updates even when its primitive-status ledger is exhausted.

The distinction is not cosmetic. A coupling value is physically meaningful, but its availability as a measured, fitted, running, or scheme-indexed value does not make it a primitive carrier. A CKM entry is physically meaningful, but its coordinate role in transition structure does not make it the primitive transition itself. A Higgs pole mass is physically meaningful, but the scalar pole-value display does not replace the Higgs bridge/fixation carrier. An EFT coefficient may be predictive, but predictive utility does not confer primitive same-scope standing.

1.4 Kernel status: non-degeneracy, not overlay

The kernel invoked by this proof is not a descriptive overlay placed on top of physics after the physical work is complete. It records the minimal necessity conditions under which physical claimhood is non-degenerate. A claim can be tested, compared, preserved, refuted, or target-updated only if it retains reference-bearing standing and a stable comparison class. Without those conditions, the terms “same object,” “same theory,” “same prediction,” “same regime,” and “same failure” no longer have determinate use.

This point is foundational for SM-6. The theorem does not add new physical entities to the Standard Model. It identifies the carrier and status conditions under which Standard Model-associated claims remain the same claims across representation, measurement, projection, anomaly pressure, and falsification. The kernel is therefore a necessity structure for non-degenerate physical reference, not a convention that may be accepted or rejected while leaving sameness intact.

1.5 Structure of the proof

The proof proceeds in nine movements. First, the formal arena D_{SM}^3 and the status grammar are fixed. Second, the dependency stack is received without re-proving the local sector papers.

Third, the global primitive carrier package is defined. Fourth, the global registered status atlas is constructed. Fifth, the open-generation cardinality endpoint is received and generation governance is instantiated at three while remaining outside the primitive carrier package. Sixth, the standalone structural spine is used to show that the global carrier package is a coherent architecture rather than a loose list. Seventh, local bridge lemmas close projection, EFT, numerical, interface, Higgs, BSM, and anomaly escape routes. Eighth, constructor adequacy, role-necessity exhaustion, and certificate-conjunct failure lemmas eliminate valid same-scope hostile-certificate closure outside the carrier package. Ninth, a global coverage theorem and no-leakage lemmas convert coverage into residual deletion.

The final result is the deletion theorem:

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset.$$

The theorem has an exposed hostile branch. A valid same-scope primitive outside C_{SM}^{prim} defeats the theorem. It is not reclassified as a skin merely because the theorem wants closure.

2 Formal arena and status grammar

2.1 The fixed domain

Definition 2.1 (Same-scope Standard Model carrier arena). D_{SM}^3 is the fixed same-scope Standard Model carrier arena under generation governance instantiated at three by the open-generation cardinality endpoint. It contains Standard Model-associated claims whose objects are asserted relative to the fixed Standard Model carrier target specified by the received structural spine: gauge order, matter representation response, charge response, coupling and renormalization transport, flavor transition, Higgs bridge/fixation, vacuum compatibility, anomaly discipline, measurement, projection, EFT display, and generation-governed indexing. It excludes claims that replace the target domain rather than occupying or describing it.

The superscript 3 records that the family-cardinality load is fixed by the separate open-generation theorem $\text{ClosedGenCert}(n) \iff n = 3$. This does not add generation cardinality to the primitive carrier package. It instantiates the governance envelope that the fixed-sector papers used without making governance a carrier.

2.2 Same-scope status

Definition 2.2 (Same-scope claim). $\text{SameScope}_{D_{SM}^3}(x)$ holds when x is asserted within the fixed carrier comparison class of D_{SM}^3 . A claim is not same-scope when it changes the domain, changes

the carrier inventory, changes the admissibility standard, changes the identity target, changes the primitive comparison class, or demands a richer target object.

Same-scope status is a discipline on comparison. It prevents equivocation between two different tasks: closing the primitive ledger of the Standard Model carrier domain and replacing the Standard Model carrier domain with a new theory. A target-updating theory may be correct; it does not supply a same-scope primitive inside D_{SM}^3 .

2.3 Kernel preservation and non-degenerate physical claimhood

Definition 2.3 (Kernel preservation). $\text{KernelPreserved}(x)$ holds when the admissibility, standing, reference, sameness, comparison-class, and failure conditions required for x to remain a determinate physical claim are preserved. $\text{NonDegeneratePhysicalClaim}(x)$ holds when x is a claim capable of target-preserving reference, comparison, prediction, or falsification.

Theorem 2.1 (Kernel necessity for physical claimhood).

$$\text{NonDegeneratePhysicalClaim}(x) \Rightarrow \text{KernelPreserved}(x).$$

Equivalently,

$$\neg \text{KernelPreserved}(x) \Rightarrow \text{DegenerateReference}(x) \vee \text{DegenerateSameness}(x) \\ \vee \text{DegenerateStanding}(x).$$

Proof. This is received from the kernel/admissibility source and instantiated in the standalone Standard Model structural spine. The dependency order is upstream-to-downstream: the kernel supplies the necessity conditions for non-degenerate claimhood, while the structural spine applies those conditions to the Standard Model carrier target. A physical claim that lacks reference-bearing standing cannot be compared to a target. A claim that lacks sameness preservation cannot be recognized as the same claim across transformation or test. A claim that lacks a fixed comparison class cannot be distinguished from target replacement. Hence non-degenerate physical claimhood requires kernel preservation. $\square$

Theorem 2.2 (Sameness requires kernel preservation).

$$\text{SameClaim}(x, y) \Rightarrow \text{KernelPreserved}(x, y).$$

Contrapositively,

$$\neg \text{KernelPreserved}(x, y) \Rightarrow \neg \text{SameClaim}(x, y).$$

Proof. Sameness is not bare verbal continuity. It requires preservation of the admissibility, target, standing, reference, and comparison conditions under which the claim is identified. If those conditions

are not preserved, the second item is a replacement, degeneration, target update, or failed comparison rather than the same claim under test. $\square$

Remark 2.1 (Reality contact). The reality contact of prediction and falsification is supplied by target-preserving reference. A prediction is a prediction of the same claim only if the target and comparison class are preserved. A falsification is a falsification of the same claim only if the failure is measured against the preserved object under test. The kernel does not insert a metaphysical layer between theory and world; it states the conditions under which theory-world comparison has a determinate object.

2.4 Standard Model-associated claims

Definition 2.4 (Standard Model-associated claim). $\text{SMClaim}(x)$ holds when x is presented as a Standard Model object, Standard Model measurement, Standard Model parameter, Standard Model route, Standard Model projection, Standard Model anomaly, Standard Model extension, or BSM claim positioned against the Standard Model target.

This definition is intentionally broad. It includes ordinary SM descriptions and adversarial BSM submissions. Breadth at the claim level is necessary for a strong endpoint: the theorem does not hide possible escapes by excluding them from the claim space.

2.5 Attribute space

Definition 2.5 (Attribute membership lock). For SM-6 the attribute arena is definitionally locked by

$$\text{Attr}(D_{SM}^3) := \{x : \text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x)\}.$$

Thus an object belongs to the global attribute arena exactly when it is asserted as a Standard Model-associated claim inside the fixed same-scope carrier domain. Claims that alter the target domain are handled by the target-update theorem and are not elements of $\text{Attr}(D_{SM}^3)$.

Definition 2.6 (Constructor presentation of the attribute arena). The same arena is presented by the following exhaustive constructors:

$$\begin{aligned} \text{Attr}(D_{SM}^3) = & \mathcal{A}_G \cup \mathcal{A}_M \cup \mathcal{A}_{CCR} \cup \mathcal{A}_F \cup \mathcal{A}_H \cup \mathcal{A}_{gen} \cup \mathcal{A}_{num} \cup \mathcal{A}_{meas} \\ & \cup \mathcal{A}_{proj/EFT} \cup \mathcal{A}_{BSM/anom} \cup \mathcal{A}_{int} \cup \mathcal{A}_{ov}. \end{aligned}$$

Here $\mathcal{A}_G$ is gauge-sector attribute generation; $\mathcal{A}_M$ is matter-sector attribute generation; $\mathcal{A}_{CCR}$ is charge, coupling, and renormalization generation; $\mathcal{A}_F$ is flavor, mixing, transition, and CP generation; $\mathcal{A}_H$ is Higgs-sector generation; $\mathcal{A}_{gen}$ is generation-cardinality and family-index generation; $\mathcal{A}_{num}$ is numerical, fit, calibration, and value generation; $\mathcal{A}_{meas}$ is measurement, detector, event, evidence,

and route generation; $\mathcal{A}_{\text{proj/EFT}}$ is projection, EFT, Wilsonian, matching, and effective-description generation; $\mathcal{A}_{\text{BSM/anom}}$ is BSM, anomaly, excess, extension, and model-surface generation; $\mathcal{A}_{\text{int}}$ is cross-sector mediation and interface generation; and $\mathcal{A}_{\text{ov}}$ records overlaps among these constructors.

Lemma 2.3 (Attribute Membership Lock).

$$x \in \text{Attr}(D_{SM}^3) \iff \text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x).$$

Proof. This is the definitional lock for the SM-6 arena. It prevents the final deletion theorem from ranging over an attribute class wider than the coverage theorem and prevents target-updating claims from being counted as same-scope survivors. $\square$

Lemma 2.4 (Constructor Exhaustion).

$$\begin{aligned} x \in \text{Attr}(D_{SM}^3) \Rightarrow & x \in \mathcal{A}_G \vee x \in \mathcal{A}_M \vee x \in \mathcal{A}_{CCR} \vee x \in \mathcal{A}_F \vee x \in \mathcal{A}_H \\ & \vee x \in \mathcal{A}_{\text{gen}} \vee x \in \mathcal{A}_{\text{num}} \vee x \in \mathcal{A}_{\text{meas}} \vee x \in \mathcal{A}_{\text{proj/EFT}} \\ & \vee x \in \mathcal{A}_{\text{BSM/anom}} \vee x \in \mathcal{A}_{\text{int}} \vee x \in \mathcal{A}_{\text{ov}}. \end{aligned}$$

Proof. The constructor presentation is a finite presentation of the global attribute arena. Since $\text{Attr}(D_{SM}^3)$ is defined as the union of these constructors, membership in the arena entails membership in at least one constructor class. Exhaustiveness is therefore not an empirical list of familiar vocabulary; it is the formal presentation of the domain to which the deletion theorem applies. $\square$

Lemma 2.5 (Constructor Adequacy).

$$\begin{aligned} \text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \Rightarrow & x \in \mathcal{A}_G \vee x \in \mathcal{A}_M \vee x \in \mathcal{A}_{CCR} \vee x \in \mathcal{A}_F \vee x \in \mathcal{A}_H \\ & \vee x \in \mathcal{A}_{\text{gen}} \vee x \in \mathcal{A}_{\text{num}} \vee x \in \mathcal{A}_{\text{meas}} \vee x \in \mathcal{A}_{\text{proj/EFT}} \\ & \vee x \in \mathcal{A}_{\text{BSM/anom}} \vee x \in \mathcal{A}_{\text{int}} \vee x \in \mathcal{A}_{\text{ov}}. \end{aligned}$$

Proof. By Attribute Membership Lock, $\text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x)$ is equivalent to $x \in \text{Attr}(D_{SM}^3)$. Constructor Exhaustion then applies. The adequacy claim is not closure by arbitrary definition: the constructors are generated by the received Standard Model carrier architecture itself—gauge, matter, charge/coupling/RG, flavor, Higgs, generation governance, numerical and measured statuses, projection/EFT displays, BSM/anomaly surfaces, cross-sector interfaces, and overlaps. A successful objection must therefore name a same-scope Standard Model-associated claim not generated by any constructor and also show that it is not a target update. Without that two-part showing, it is either inside the constructor presentation or outside D_{SM}^3 . $\square$

Lemma 2.6 (Official-Target Constructor Adequacy).

$$\begin{aligned} & \text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \wedge x \notin \mathcal{A}_G \wedge x \notin \mathcal{A}_M \wedge x \notin \mathcal{A}_{CCR} \wedge x \notin \mathcal{A}_F \wedge x \notin \mathcal{A}_H \\ & \wedge x \notin \mathcal{A}_{gen} \wedge x \notin \mathcal{A}_{num} \wedge x \notin \mathcal{A}_{meas} \wedge x \notin \mathcal{A}_{proj/EFT} \\ & \wedge x \notin \mathcal{A}_{BSM/anom} \wedge x \notin \mathcal{A}_{int} \wedge x \notin \mathcal{A}_{ov} \Rightarrow \perp. \end{aligned}$$

Proof. The constructor list is generated by the received Standard Model target rather than by an ad hoc vocabulary choice. A same-scope Standard Model-associated attribute must arise through gauge standing, matter representation response, charge/coupling/RG transport, flavor transition, Higgs bridge/fixation, generation governance, numerical or measurement status, projection/EFT display, BSM/anomaly surface, cross-sector interface, or an overlap of these constructors. A missing-constructor objection must exhibit a same-scope SM-associated attribute that is not gauge, not matter, not CCR, not flavor, not Higgs, not generation-governed, not numerical or measured, not projection/EFT, not BSM/anomaly, not interface/overlap, and not target-updating. Such an object is not an element of the received SM target. If it is scientifically meaningful, it belongs to a target update or to a separate closure problem. $\square$

Proposition 2.7 (Structural-Spine Constructor Generation). *Relative to the received kernel and standalone Standard Model structural spine, the ordinary Standard Model carrier target generates exactly the constructor families used above:*

$$\begin{aligned} \text{Spine}_{SM} \Rightarrow & \mathcal{A}_G \cup \mathcal{A}_M \cup \mathcal{A}_{CCR} \cup \mathcal{A}_F \cup \mathcal{A}_H \cup \mathcal{A}_{gen} \cup \mathcal{A}_{num} \cup \mathcal{A}_{meas} \cup \\ & \mathcal{A}_{proj/EFT} \cup \mathcal{A}_{BSM/anom} \cup \mathcal{A}_{int} \cup \mathcal{A}_{ov}. \end{aligned}$$

Proof. The structural spine supplies the Standard Model-specific architecture: gauge product, matter representation response, charge/coupling/RG transport, Higgs bridge/fixation, flavor transition, generation governance, anomaly compatibility, vacuum compatibility, and parameter-status separation. Measurement records, numerical values, EFT projections, BSM/anomaly surfaces, and cross-sector interfaces are then the ordinary downstream status families generated by applying that architecture to prediction, calibration, comparison, and extension claims. Hence the constructor list is not a private taxonomy detached from ordinary physics vocabulary; it is the finite carrier/status presentation of the received Standard Model target under the kernel conditions for non-degenerate sameness and reference. $\square$

2.6 Standard-physics correspondence of the constructors

The constructor presentation is a formal carrier-status presentation of recognizable Standard Model material. It is not a private taxonomy detached from ordinary physics vocabulary.

Ordinary Standard Model object or status	Constructor class	SM-6 status
$SU(3) \times SU(2) \times U(1)$ gauge structure	$\mathcal{A}_G$	gauge carrier standing
Gauge-boson labels and field symbols	$\mathcal{A}_G$	representative labels descending to gauge carriers
Quark representations	$\mathcal{A}_M$	color-bearing matter standing
Charged leptons	$\mathcal{A}_M$	colorless charged matter standing
Neutrinos	$\mathcal{A}_M$	neutral weak matter standing
Electric charge and hypercharge	$\mathcal{A}_{CCR}$	charge-response and charge-lattice standing
g_1, g_2, g_3, e, α	$\mathcal{A}_{CCR}, \mathcal{A}_{num}$	coupling modulus or calibration status
RG running and thresholds	$\mathcal{A}_{CCR}, \mathcal{A}_{\text{proj/EFT}}$	transport, scheme, or matching status
CKM matrix	$\mathcal{A}_F$	transition-coordinate status
PMNS matrix	$\mathcal{A}_F$	boundary-reconstruction or transition-coordinate status
CP phase content	$\mathcal{A}_F$	CP/holonomy residue or phase-coordinate status
Higgs scalar notation	$\mathcal{A}_H$	representative skin descending to $H_{br/fix}$
Higgs vev	$\mathcal{A}_H, \mathcal{A}_{num}$	fixation, route, or value status
Higgs potential	$\mathcal{A}_H$	route or representative status
Particle masses and widths	$\mathcal{A}_{num}$	measured response, fixation, or pole-value status
Detector events and excesses	$\mathcal{A}_{meas}$	evidence, measurement, route, or anomaly-pressure status
EFT operators	$\mathcal{A}_{\text{proj/EFT}}$	projection or matching status
Wilson coefficients	$\mathcal{A}_{\text{proj/EFT}}, \mathcal{A}_{num}$	coefficient/projection status
Anomaly reports	$\mathcal{A}_{\text{BSM/anom}}, \mathcal{A}_{meas}$	evidence pressure, repair demand, burden, failure, or target update
BSM model surfaces	$\mathcal{A}_{\text{BSM/anom}}$	extension, burden, failure, hostile attempt, or target update

2.7 Primitive standing

Definition 2.7 (Primitive standing). $\text{Prim}_D(x)$ holds when x is an irreducible carrier ground inside domain D , not merely a notation, coordinate, measurement value, route, projection, scheme artifact, calibration status, representative, extension surface, burden, failed primitive attempt, or target update.

Primitive standing is role-theoretic. It is not conferred by familiarity, numerical precision, empirical salience, explanatory utility, or frequent use in calculations. Those features can make a skin indispensable without making it primitive.

2.8 Primitive-survivor operator

Definition 2.8 (Primitive-survivor operator). For any domain D and any attribute set S ,

$$\text{Survprim}_D(S) := \{x \in S : \text{Prim}_D(x)\}.$$

The notation $\text{Survprim}_D(S) = 0$ abbreviates the assertion that the survivor set is empty, equivalently $\text{Survprim}_D(S) = \emptyset$ or $|\text{Survprim}_D(S)| = 0$. In SM-6 the carrier exclusion is written directly in the argument, $S = \text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}$. Registered, governed, burden, failed, and target-updating cases are eliminated by the status, no-leakage, and certificate-failure theorems before this equality is concluded.

2.9 Descent

Definition 2.9 (Descent). $x \downarrow C$ means that x is licensed by, realized through, measured relative to, transported along, projected from, coordinated by, or otherwise status-dependent on the carrier package C . Descent is not identity.

For example, a CKM entry descends to the flavor-transition carrier package as a transition coordinate. A running gauge coupling descends to charge/coupling/renormalization transport as a scheme-indexed modulus. A Higgs pole mass descends to $H_{br/fix}$ as a measured scalar/bridge status. None of these descents identify the downstream object with the primitive carrier itself.

Lemma 2.8 (Carrier-descent incompatibility).

$$x \downarrow C_{SM}^{\text{prim}} \Rightarrow x \notin \text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}).$$

Equivalently, within the residual-survivor test, carrier descent is incompatible with irreducible primitive survival outside the carrier package.

Proof. Descent means that the claim is licensed by, measured relative to, projected from, transported along, or otherwise status-dependent on the certified carrier package. Such dependence is the denial of irreducible carrier ground for the descended claim. The descended object may be physically real, measurable, predictive, or indispensable in calculation; it simply does not supply independent primitive carrier standing outside C_{SM}^{prim} . $\square$

2.10 Registered status

Definition 2.10 (Registered nonprimitive status atlas). T_{SM}^{reg} is the global registered atlas of nonprimitive Standard Model statuses. It contains symbolic representatives, particle labels, scalar and spinor notations, parameter values, coupling moduli, running values, scheme coordinates, matching thresholds, EFT coefficients, CKM and PMNS coordinates, phase coordinates, basis choices, detector records, routes, projections, BSM model surfaces, anomaly statuses, extension dispositions, and evidence statuses.

The registry is open to refinement as a classification atlas. It is closed against primitive promotion except through a valid hostile same-scope certificate. A new row in the registry is not a new primitive. It is a new or refined nonprimitive status.

2.11 Generation governance

Definition 2.11 (Generation governance). $\text{GenGoverned}_3(x)$ holds when x is licensed, indexed, or classified under fixed generation-cardinality governance G_{gen}^3 , where the value three is supplied by the open-generation endpoint. Governance is not primitive carrier standing.

Generation governance is retained because matter and flavor closure operate under fixed family load. The cardinality value is now received from D_{gen} , while governance remains outside C_{SM}^{prim} to avoid smuggling a generation primitive into the carrier set.

2.12 Burden, failure, hostile branch, target update

Definition 2.12 (Counterexample burden). $\text{CounterexampleBurden}_{SM}(x)$ holds when x asserts or gestures toward same-scope primitive standing outside C_{SM}^{prim} but has not supplied a valid same-scope role-occupancy certificate.

Definition 2.13 (Failed primitive). $\text{FailedPrimitive}_{SM}(x)$ holds when x attempts primitive standing but collapses into registered status, target update, inconsistency, role insufficiency, representative status, projection status, or derivative descent.

Definition 2.14 (Hostile certificate branch). $\text{HostileCertificateBranch}_{SM}(x)$ holds when x is submitted as a same-scope, nonderivative, nonregistered, nongovernance, non-target-updating primitive outside C_{SM}^{prim} , with the burden of demonstrating role-occupancy necessity.

Definition 2.15 (Target update). $\text{TargetUpdate}_{SM}(x)$ holds when x alters the target domain, carrier class, admissibility regime, primitive comparison standard, or identity object rather than occupying the fixed same-scope Standard Model arena.

The hostile branch is not a disposal bin. If a hostile certificate closes, SM-6 is false. The classifier is therefore not immunizing itself against counterexample.

Definition 2.16 (Valid hostile certificate). A valid hostile certificate is the positive same-scope role-occupancy predicate

$$\begin{aligned} \text{ValidHostileCertificate}_{SM}(x) \iff & x \in \text{Attr}(D_{SM}^3) \wedge x \notin C_{SM}^{\text{prim}} \wedge \text{Prim}_{D_{SM}^3}(x) \\ & \wedge \neg(x \downarrow C_{SM}^{\text{prim}}) \wedge x \notin T_{SM}^{\text{reg}} \\ & \wedge \neg \text{GenGoverned}_3(x) \wedge \neg \text{TargetUpdate}_{SM}(x) \\ & \wedge \text{RoleNecessity}_{SM}(x) \wedge \text{FixedComparisonClass}_{SM}(x). \end{aligned}$$

A hostile-certificate branch is a certificate attempt. It is valid only if all conjuncts above close.

Lemma 2.9 (Fixed-comparison failure).

$$\neg \text{FixedComparisonClass}_{SM}(x) \Rightarrow \text{TargetUpdate}_{SM}(x) \vee \text{FailedPrimitive}_{SM}(x).$$

Proof. The fixed comparison class is the condition that the submitted primitive role is evaluated inside the same Standard Model carrier target. If that class is changed, the claim updates the target. If the claim denies the fixed comparison class while still presenting itself as a same-scope primitive, the primitive attempt fails by equivocation. Thus fixed-comparison failure is not an internal survivor branch. $\square$

Remark 2.2 (Falsifiability and no-valid-certificate closure). Relative to the received dependency stack and the constructor-presented arena D_{SM}^3 , the theorem proves that no valid hostile certificate closes. A future or external submission that appears to close such a certificate is not absorbed into the classifier. It attacks at least one load-bearing item: a sector closure, the Higgs endpoint, the structural-spine embedding, constructor adequacy, role-necessity exhaustion, the target/same-scope boundary, or a bridge lemma. If that attack succeeds, SM-6 is false at the indicated point. This formulation does not immunize SM-6 against counterexample; it fixes the burden required for an internal same-scope primitive survivor.

3 Source reception and dependency discipline

3.1 Reception principle

SM-6 is a synthesis theorem. It receives closed sector packages and assembles them into a global endpoint. It does not re-prove the local results of SM-1 through SM-5 or the Higgs endpoint papers. A valid attack on SM-6 may break an imported theorem, break a domain embedding, break a bridge lemma proved here, supply a faithful same-scope hostile certificate, or show that a target-update boundary has been misapplied. A valid attack is not created merely by observing that SM-6 depends on the upstream closure sequence.

Reception 3.1 (Dependency-lock posture). A received theorem is used at its declared output. Dependence is not a defect; a successful challenge must break a load-bearing dependency or supply a faithful same-scope counterexample.

3.2 Kernel/admissibility import: non-degeneracy source

The kernel/admissibility source, *Non-Degenerate Construction and the Kernel of Admissibility*, supplies the upstream necessity claim used throughout SM-6. It is not received from the Standard Model structural spine. Rather, the structural spine instantiates it in the Standard Model target. The imported kernel result is:

$$\text{NonDegeneratePhysicalClaim}(x) \Rightarrow \text{KernelPreserved}(x),$$

with the corresponding sameness, standing, reference, and comparison-class preservation requirements. The SM-specific carrier architecture is downstream of this kernel; the kernel is not downstream of the architecture.

3.3 SM-1 import: classifier-level discipline

The first source, *Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model*, supplies the master classifier. SM-1 fixes the claim arena, the carrier/skin distinction, target-update discipline, future-claim classifiability, burden status, prediction register, falsification protocol, BSM/EFT branch logic, and handoff architecture.

SM-1 is not imported as a proof of global primitive exhaustion. It supplies classifier discipline: skins do not become primitive merely by utility, visibility, or empirical refinement. Future claims must either descend to a carrier package, enter a registered status, bear a burden, fail as primitives, expose a hostile branch, or update the target.

3.4 SM-2 import: gauge carriers

The second source, *Gauge Carriers: Photon, Gluon, and Weak-Boson Standing*, supplies the gauge-sector carrier closure. The imported carrier package is

$$C_{\text{gauge}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly-compat}}, C_{\text{gauge-vacuum}}\}.$$

The imported theorem is

$$\text{Survprim}_{D_{\text{gauge}}}(\text{Attr}_G \setminus C_{\text{gauge}}) = 0.$$

This closes gauge primitive-status escape for photon, gluon, weak-boson, curvature, holonomy, transport, anomaly-compatibility, gauge-vacuum, charge-response, color-transport, and weak-chiral standing at the local gauge-sector level. The notation $C_{\text{anomaly-compat}}$ names the structural anomaly-compatibility role; it is distinct from empirical anomaly reports, which enter T_{SM}^{reg} as evidence pressure, burden, failure, or target-update triggers.

3.5 SM-3 import: matter carriers

The third source, *Matter Carriers: Quarks, Leptons, and Neutrinos*, supplies matter-sector closure under fixed generation governance. The imported carrier package is

$$C_{\text{matter}} = \{C_{\text{matter-rep}}, C_{\text{color-bearing-matter}}, \\ C_{\text{colorless-charged-matter}}, C_{\text{neutral-weak-matter}}, \\ C_{\text{mass-response-slot}}\}.$$

The imported theorem is

$$\text{Survprim}_{D_{\text{matter}}^3}(\text{Attr}(D_{\text{matter}}^3) \setminus C_{\text{matter}}) = 0.$$

Detector tracks, jets, hadron membership, oscillation routes, cosmological bounds, and flavor labels are imported as derivative, route, projection, or governance-related statuses rather than primitive matter standing.

3.6 SM-4 import: charges, couplings, and renormalization

The fourth source, *Charges, Couplings, and Renormalization as Standing Transport*, supplies the charge/coupling/renormalization closure. The primitive package is

$$C_{CCR}^{\text{prim}} = \{C_{\text{charge-response}}, C_{\text{charge-witness}}, C_{\text{charge-lattice}}\}.$$

The registered CCR status package includes coupling moduli, standing-preserving coupling status, RG transport, scheme and matching status, calibration values, EW/EM bookkeeping, and EFT/projection status. The imported theorem is

$$\text{Survprim}_{D_{CCR}}(\text{Attr}(D_{CCR}) \setminus C_{CCR}^{\text{prim}}) = 0.$$

Numerical coupling values, running curves, threshold matchings, fixed points, unification plots, and scheme coordinates are imported as nonprimitive statuses.

3.7 SM-5 import: flavor, mixing, and generation

The fifth source, *Flavor Transition, Mixing, and Generation Cardinality*, supplies flavor/mixing closure under fixed generation governance. The primitive package is

$$C_{\text{flavor}}^{\text{prim}} = \{C_{\text{flavor-transition}}, C_{CP/\text{holonomy-residue}}\}.$$

The imported theorem is

$$\text{Survprim}_{D_{\text{flavor}}^3}(\text{Attr}(D_{\text{flavor}}^3) \setminus C_{\text{flavor}}^{\text{prim}}) = 0.$$

CKM entries, PMNS entries, mixing angles, raw phase coordinates, basis choices, rephasing conventions, diagonalization, mass/flavor eigenstate labels, oscillation routes, FCNC claims, flavor anomalies, fourth-generation claims, and flavor-sector BSM surfaces are imported as transition coordinates, boundary reconstructions, routes, burden branches, target updates, failed primitives, or hostile-certificate submissions. A same-scope role certificate must close independently.

3.8 Open-generation cardinality endpoint import

The primitive-standing reference edition of *Generation Cardinality as a Forced Role-Occupancy Endpoint* [R6] supplies

$$\text{Prim}_{\mathcal{P}_0}(I) \iff |I| = 3, \quad \text{ClosedGenCert}_{\mathcal{P}_0}(n) \iff n = 3.$$

Here primitive standing is joint constructibility of the independently specified original work with no proper adequate occurrence support. The source's Theorem 5.1 supplies all duties jointly on every live triangle; Theorem 6.2 classifies primitive architectures, and Theorem 5.4 counts complete occupation of their full live union. Complete comparison and vertex rephasing exclude smaller architectures. Every larger adequate architecture has a proper triple adequate for the same work and therefore lacks primitive standing. Corollary 5.5 derives rank one only after the architecture is established. Empirical family labels, matrix dimension, fitted textures, and a chosen orientation are

not count selectors.

The inherited representation, charge, weak and bridge licenses come from the sector sources. Their compatibility is received through the source's common packet, not inferred merely from writing matrices. The following map states how that packet enters the SM-6 governance object.

Definition 3.1 (Primitive generation-governance receipt). A receipt consists of a primitive $\mathcal{P}_0$ construction p , its homogeneous occurrence architecture $R(p)$, a claimed primitive occurrence set X , and a joint injective assignment $\sigma : R(p) \rightarrow X$ covering the full live occurrence set. It retains the actual role eligibility, occurrence incidence, comparison quotient, boundary domains and outcomes, and lawful assignment action. Original primitive readers are interpreted by their receiving expressions; the count is not a field fixed in advance.

Proposition 3.1 (The SM-6 primitive governance interpretation). *A receipt of Definition 3.1 has $|X| = 3$ and therefore supplies the governance object G_{gen}^3 . Such receipts exist within the received Standard Model type/bridge/boundary scope. They add no member to the primitive carrier package C_{SM}^{prim} .*

Proof. The source's Theorem 6.2 gives $|R(p)| = 3$, and the complete injective assignment is a bijection onto X . For a positive receipt, take its Proposition 4.4 packet. Set X equal to the rank-one orthogonal projections commuting with its simple charged response A . Spectral simplicity makes these exactly the intrinsic occurrence projections P_i . Assign each homogeneous position to its P_i ; this is a joint and exhaustive assignment. Pair readers $\text{Tr}(P_i B P_j B)$ and triangle readers $\text{Tr}(P_i B P_j B P_k B)$ receive the complete comparison and its phase quotient. The same common packet receives the inherited gauge, charge, weak, mass-slot, bridge, neutral-boundary, and failure data. Its Theorem 4.2 and Proposition 4.5 give joint compatibility and typed covariance of the declared generated uses. Reindexing the homogeneous positions transports the queries, leaving the physical responses fixed.

Shared charged-sector multiplicity is separately justified by the received representation/hypercharge inventory and anomaly equations [R6, Proposition 4.1]. It does not identify quark and lepton mass eigenlines; nor does it constrain additional hypercharge-zero singlet multiplicity. The construction thus receives a count and an occupation relation for the original generation work. It does not create an additional physical carrier role beside the gauge, matter, CCR, flavor, and Higgs roles. $\square$

The map just constructed supplies primitive-governance data from the shared packet. A different physical recipient must retain the same original primitive meanings, or provide a nonempty licensed receipt relation. The source's Theorem 6.6 makes its primitive count constant without selecting a preferred embedded triangle; Proposition 6.7 requires agreement on each receipt fibre for any other original reader claimed to descend. The complete enriched response algebra is not silently identified with this primitive interpretation. Governance is instantiated at G_{gen}^3 while remaining outside C_{SM}^{prim} .

3.9 Higgs endpoint import

The Higgs capstone, *The Higgs Domain Endpoint: Carrier, Identity, Scale, Vacuum, Extension, and Projection*, supplies the Higgs carrier endpoint

$$C_H = H_{br/fix}.$$

Its support stack includes the Higgs structural identity endpoint, the no-primitive-Higgs-scale theorem, the Higgs extension classifier, vacuum-route standing discipline, lower-vacuum default discipline, fixed-domain exhaustion, closure by exhaustion, kernel/admissibility supports, and ametric boundary supports.

SM-6 imports this package as closing same-scope primitive Higgs escape. Scalar notation, vev presentation, potential form, pole mass, width, hierarchy comparison, metastability, lower-vacuum claims, extension-scalar claims, and metric projection enter as representatives, routes, measurement statuses, comparison statuses, extension dispositions, burdens, target updates, or hostile branches relative to $H_{br/fix}$.

3.10 Standalone structural spine import

The source *The Standalone Standard Model Structure from First Principles: An AASC Derivation of Gauge-Product, Representation, Higgs/Bridge, Yukawa, Charge, Anomaly, Vacuum, and Parameter-Status Structure* supplies the Standard Model-specific structural spine. It links gauge-product structure, representation-response structure, Higgs/bridge structure, Yukawa realizability, charge lattice, anomaly compatibility, vacuum compatibility, and parameter-status separation. It does not originate the kernel necessity claim; it instantiates the upstream kernel/admissibility requirements in the Standard Model target. Its role in SM-6 is to prevent the final carrier package from reading as a disjoint catalogue and to generate the constructor presentation of $\text{Attr}(D_{SM}^3)$. The global package is one carrier architecture.

4 The global primitive carrier package

4.1 Definition

Definition 4.1 (Global primitive carrier package). The primitive carrier package of SM-6 is

$$C_{SM}^{\text{prim}} = C_{\text{gauge}} \cup C_{\text{matter}} \cup C_{CCR}^{\text{prim}} \cup C_{\text{flavor}}^{\text{prim}} \cup \{H_{br/fix}\}.$$

Expanded:

$$C_{SM}^{\text{prim}} = \{C_{\text{gauge-order}}, C_{\text{transport}}, C_{\text{curv/hol}}, C_{\text{charge-response}}, \\ C_{\text{color-transport}}, C_{\text{weak-chiral}}, C_{\text{anomaly-compat}}, C_{\text{gauge-vacuum}}, \\ C_{\text{matter-rep}}, C_{\text{color-bearing-matter}}, C_{\text{colorless-charged-matter}}, \\ C_{\text{neutral-weak-matter}}, C_{\text{mass-response-slot}}, \\ C_{\text{charge-witness}}, C_{\text{charge-lattice}}, C_{\text{flavor-transition}}, \\ C_{\text{CP/holonomy-residue}}, H_{\text{br/fix}}\}.$$

The repeated occurrence of $C_{\text{charge-response}}$ in the gauge and CCR imports is treated as a shared role, not a multiplication of primitive carriers.

Lemma 4.1 (Shared-role coherence). *If a carrier role occurs in two received sector packages, its global occurrence is an interface-identical shared carrier role. Distinct primitive occupancy requires explicit distinction by the receiving theorem.*

Proof. A sector package is received by role standing, not by textual occurrence. When the same role name and function mediate the same standing relation across two sectors, duplicating it as two primitive carriers would require an additional distinguishing certificate. No such certificate is supplied or needed for $C_{\text{charge-response}}$. The role is therefore shared. Shared occurrence preserves carrier standing and avoids primitive multiplication by notation. $\square$

Lemma 4.2 (Union does not generate primitives). *The formation of C_{SM}^{prim} generates no new primitive carrier merely by taking the union of received packages.*

Proof. The union operation records carriers already licensed by local sector closure or the Higgs endpoint. Any object that appears only as an interface, translation, projection, bookkeeping device, scheme choice, or cross-sector display is not thereby admitted into C_{SM}^{prim} . Primitive status requires a carrier certificate. Union alone is not such a certificate. $\square$

Proposition 4.3 (Sufficient carrier package). *C_{SM}^{prim} is sufficient for the global same-scope primitive-status closure proved in this paper.*

Proof. Sufficiency follows from the received local deletion theorems, the Higgs endpoint, the structural-spine coherence theorem, the sector-embedding theorem, the projection/interface lemmas, the BSM/EFT audit theorem, the global coverage theorem, and the no-leakage theorem proved below. The package need not be shown irredundantly minimal for sufficiency. It is enough that no attribute outside it survives as primitive. $\square$

5 The global registered status atlas

5.1 Definition of T_{SM}^{reg}

The registered atlas T_{SM}^{reg} contains nonprimitive statuses that remain legitimate Standard Model discourse. The following table is a binding registry for the proof, not an exhaustive list of every possible notation.

Status class	Examples	Primitive-status ruling
Symbolic representative	field symbols, scalar notation, spinor notation, gauge-field notation, particle names	licensed presentation, not carrier ground
Particle label	photon, gluon, weak boson, quark, lepton, neutrino, Higgs label	label over carrier roles, not primitive by naming
Numerical measurement	pole masses, widths, lifetimes, branching ratios, fit values	measurement/calibration status
Coupling modulus	g_1, g_2, g_3, e, α , weak mixing angle	modulus, transport, calibration
RG transport	beta displays, running curves, thresholds, schemes	transport and scheme status
Flavor coordinate	CKM entries, PMNS entries, mixing angles	transition coordinate or boundary reconstruction
Phase coordinate	raw CP phases, rephasing conventions	coordinate skin; holonomy residue is certified separately
Higgs representative	vev, potential, scalar field notation, pole mass	bridge/fixation representative or measurement status
Vacuum route	metastability, lower branch, thermal route, tunneling route	route/comparison status
EFT/projection	operators, Wilson coefficients, matching surfaces	projection or coefficient status
Detector/evidence	events, excesses, tracks, jets, cosmological bounds	evidence or route status
BSM surface	extra scalar, sterile state, dark sector, unification model, composite model	extension surface, burden, target update, or hostile branch

Anomaly status	flavor anomaly, detector excess, precision tension	evidence pressure, repair demand, burden, target update, or hostile branch
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Theorem 5.1 (Global status registration). *For every same-scope Standard Model-associated attribute $x \notin C_{SM}^{\text{prim}}$, if x is a familiar Standard Model description, value, route, projection, coordinate, measurement, extension surface, anomaly report, or BSM audit object, then x has a registered nonprimitive status, generation-governance status, burden status, failure status, hostile-certificate status, or target-update boundary.*

Proof. The claim classes listed above exhaust the ordinary noncarrier vocabulary of the Standard Model target and the adversarial extensions positioned against it. Gauge descriptions outside C_{gauge} are handled by SM-2. Matter descriptions outside C_{matter} are handled by SM-3. CCR descriptions outside C_{CCR}^{prim} are handled by SM-4. Flavor descriptions outside $C_{\text{flavor}}^{\text{prim}}$ are handled by SM-5. Higgs descriptions outside $H_{br/fix}$ are handled by the Higgs endpoint. Cross-sector, projection, EFT, numerical, and BSM descriptions are handled by the local SM-6 bridge theorems below. The only remaining possibility is a hostile certificate branch, which is explicitly exposed rather than registered as harmless. $\square$

Theorem 5.2 (No registered-status promotion). *For every x ,*

$$x \in T_{SM}^{\text{reg}} \wedge \neg \text{HostileCertificateBranch}_{SM}(x) \Rightarrow \neg \text{Prim}_{D_{SM}^3}(x).$$

Proof. Membership in T_{SM}^{reg} is assigned exactly when a claim is a representative, value, coordinate, route, projection, measurement, extension surface, anomaly status, evidence status, or bookkeeping status. Each such status is defined by dependence on an admitted carrier, comparison class, route, measurement context, or target boundary. Dependence is incompatible with primitive carrier standing except by defeating the registered assignment through an independent same-scope role certificate. That defeat is the hostile branch, not registry promotion. $\square$

6 Generation governance

6.1 Governance object

Generation cardinality is retained as fixed governance instantiated by the open-generation endpoint:

$$G_{\text{gen}}^3.$$

It is neither inserted into C_{SM}^{prim} nor treated as a primitive carrier. The closure domain is D_{SM}^3 , obtained by receiving $\text{ClosedGenCert}(n) \iff n = 3$ from D_{gen} . The family count is therefore not a

free primitive variable inside SM-6 and not an empirical selector.

Theorem 6.1 (Generation governance separation).

$$\text{GenGoverned}_3(x) \Rightarrow x \notin C_{SM}^{\text{prim}}.$$

Moreover, generation-governed status alone does not imply $\text{Prim}_{D_{SM}^3}(x)$.

Proof. The carrier package lists gauge, matter, CCR, flavor, and Higgs primitive roles. Generation governance is a load condition on matter and flavor claim domains. It indexes and governs families; it does not supply an independent same-scope carrier ground. The value three is forced by the open-generation role-occupancy theorem, while the governance object remains outside C_{SM}^{prim} . Generation-governed labels cannot be promoted without a hostile certificate. $\square$

Theorem 6.2 (Fourth-generation branch classification). *A fourth-generation claim is classified as a governance-branch claim, extension surface, burden, failed primitive attempt, target update, or hostile-certificate branch. It is not automatically a primitive generation carrier inside D_{SM}^3 .*

Proof. A proposed fourth simultaneous primitive occupant is tested against the unchanged original generation work and the complete receipt in Proposition 3.1. If its larger architecture fails an original duty, primitive standing already fails. If it realizes all the duties, a proper triple admits their joint construction and the larger architecture fails primitive necessity [R6, Theorems 6.2 and 6.4]. The full live-union occupation theorem separately excludes $\text{ClosedGenCert}_{\mathcal{P}_0}(4)$. Its extra cycle rank is consequently a diagnostic of enlargement, not the premise from which three is inferred. Additional physical data can retain standing for an enriched task. An indispensable original duty alleged to be missing instead challenges the original-work receipt and must be examined on that basis; it is not eliminated by calling it enrichment. A change of the target is evaluated as an extension or target update, and a genuine failure of the stated receipt or theorem hypotheses remains a hostile certificate. $\square$

7 Structural spine: one carrier architecture

7.1 Gauge-product and representation-response compatibility

The standalone structural spine establishes that the received sectors are not independent vocabularies placed side by side. Gauge order supplies admissible gauge-product standing. Matter roles occupy representation-response positions under that order. Charge response and charge witness organize how matter and gauge standing become physically addressable. Color-bearing matter, colorless charged matter, and neutral weak matter are not free particle labels; they are matter-response roles under the gauge and charge architecture.

7.2 Higgs/bridge compatibility

The Higgs bridge/fixation carrier $H_{br/fix}$ supplies the bridge by which mass-response and fixation statuses are licensed. It is not a scalar notation promoted by familiarity. Yukawa realizability, mass slots, vev presentation, and scalar-potential presentation are downstream of the bridge/fixation role. If a Yukawa object is submitted as a primitive, it is classified as bridge/fixation realization, mass-response route, flavor-transition coordinate, numerical parameter status, or hostile-certificate attempt.

7.3 Anomaly and vacuum compatibility

Anomaly and vacuum roles enter as structural admissibility and compatibility constraints. They are not mere empirical afterthoughts and not free primitive additions. A vacuum description may be a route, a branch, a comparison, or a target update. An anomaly claim may be evidence pressure or a burden branch. Neither becomes primitive without role-occupancy certification.

Theorem 7.1 (Carrier architecture coherence). *The global carrier package C_{SM}^{prim} forms a coherent Standard Model carrier architecture under the structural spine.*

Proof. The structural spine links gauge product, representation response, charge response, Higgs bridge/fixation, Yukawa realizability, flavor transition, anomaly compatibility, and vacuum compatibility into one admissible architecture. Each member of C_{SM}^{prim} occupies a role in that architecture. The union is therefore not a catalogue of names but a structured set of carrier roles whose interfaces are governed by the spine. $\square$

8 Sector reception theorem

8.1 Domain embeddings

The local domains embed into the global arena as claim-domain inclusions:

$$D_{\text{gauge}} \hookrightarrow D_{SM}^3, \quad D_{\text{matter}}^3 \hookrightarrow D_{SM}^3, \quad D_{CCR} \hookrightarrow D_{SM}^3, \quad D_{\text{flavor}}^3 \hookrightarrow D_{SM}^3, \quad D_H \hookrightarrow D_{SM}^3.$$

These embeddings do not assert numerical derivations. They state that local sector claims are same-scope Standard Model claims when interpreted inside the global carrier arena.

Lemma 8.1 (Sector residual deletion preservation). *If a received sector theorem proves*

$$\text{Survprim}_{D_i}(\text{Attr}(D_i) \setminus C_i) = 0$$

and $D_i \hookrightarrow D_{SM}^3$ as a same-scope claim-domain inclusion, then the corresponding local residue does not reappear as primitive in D_{SM}^3 . A submitted cross-sector primitive is handled by the hostile-certificate protocol rather than by silent absorption.

Proof. Embedding a local same-scope claim into D_{SM}^3 adds global context; it does not erase the local status ruling. A local derivative, route, coordinate, projection, or burden status remains nonprimitive when placed in a larger same-scope arena. Primitive reappearance would require a new cross-sector role not present in the local closure. That is exactly the hostile-certificate branch. Without such a certificate, local deletion is preserved. $\square$

Definition 8.1 (Cross-sector hostile certificate). A cross-sector hostile certificate is a claim x satisfying all of the following: x is same-scope inside D_{SM}^3 ; $x \notin C_{SM}^{\text{prim}}$; $x \notin T_{SM}^{\text{reg}}$; x is not generation-governed; x is not a target update; and x demonstrates role-occupancy necessity as a primitive carrier.

Lemma 8.2 (Cross-sector certificate exposure). A valid cross-sector hostile certificate falsifies the global deletion theorem. It is not absorbed by the sector union.

Proof. The deletion theorem asserts no primitive survivor outside C_{SM}^{prim} . A valid cross-sector certificate is precisely a primitive survivor outside C_{SM}^{prim} . Treating it as absorbed would contradict the theorem's own falsification condition. Therefore the branch is exposed. $\square$

9 Projection, interface, and EFT discipline

9.1 Projection-interface problem

The global synthesis uses projections, interfaces, comparisons, measurements, reductions, and matching maps. A hostile objection may claim that these interfaces introduce a new primitive. The objection succeeds only if the interface itself carries irreducible same-scope standing not already licensed by the carrier package. Mediation, display, comparison, and matching are not role necessity by themselves.

Lemma 9.1 (Projection interfaces do not promote by mediation). If p is a projection, interface, translation, display, measurement route, EFT reduction, matching surface, or coordinate comparison between already certified carrier roles, then

$$\text{ProjectionInterface}_{SM}(p) \wedge \text{Mediates}(p, C_{SM}^{\text{prim}}) \wedge \neg \text{HostileCertificateBranch}_{SM}(p) \Rightarrow p \in T_{SM}^{\text{reg}}.$$

In particular, mediation alone does not place p in C_{SM}^{prim} .

Proof. A projection or interface is typed by its function: it displays, transports, compares, measures, reduces, or translates. These functions presuppose the carrier roles between which the projection operates. They do not by themselves supply a new irreducible carrier ground. If an interface is claimed to exceed mediation, the excess claim must be submitted as a hostile certificate attempt. Otherwise it is registered. $\square$

Lemma 9.2 (Projection role-failure).

$$\begin{aligned} & \text{ProjectionInterface}_{SM}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \wedge \neg(x \downarrow C_{SM}^{\text{prim}}) \\ & \Rightarrow x \in T_{SM}^{\text{reg}} \vee \text{CounterexampleBurden}_{SM}(x) \\ & \quad \vee \text{FailedPrimitive}_{SM}(x) \vee \text{HostileCertificateBranch}_{SM}(x). \end{aligned}$$

Moreover, a mere projection-interface claim fails $\text{RoleNecessity}_{SM}(x)$ while it remains merely a projection. If it identifies a certified carrier already in C_{SM}^{prim} , carrier descent applies; if it changes the target, the target-update boundary applies.

Proof. A same-scope projection that does not descend to the carrier package is still a projection only if its role is display, comparison, translation, matching, or measurement route. These are registered statuses. If the claim asserts more, it bears a burden or fails when the alleged primitive collapses back into projection function. The only remaining branch is a hostile certificate attempt. That attempt cannot close while the object remains merely a projection, because the role supplied is mediation rather than necessity for same-scope Standard Model standing. If the alleged role is necessary, it is either a carrier already in C_{SM}^{prim} or a target update. $\square$

Theorem 9.3 (EFT display classification). *An EFT object associated with the Standard Model is a projection display, matching status, coefficient status, route status, extension surface, burden branch, target update, or hostile-certificate branch. It is not primitive merely by being predictive, useful, calibrated, or systematic.*

Proof. EFT objects are organized by projection from a target, expansion relative to a regime, matching across thresholds, and coefficient assignment. These are registered status functions. If an EFT object merely records a low-energy or high-energy display, it is a projection. If it introduces a new target, it is a target update. If it asserts a same-scope primitive outside C_{SM}^{prim} , it bears the hostile-certificate burden. No other route to primitive status is licensed. $\square$

Lemma 9.4 (EFT role-failure).

$$\begin{aligned} & \text{EFTClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \wedge x \notin T_{SM}^{\text{reg}} \\ & \Rightarrow \text{CounterexampleBurden}_{SM}(x) \vee \text{FailedPrimitive}_{SM}(x) \vee \text{HostileCertificateBranch}_{SM}(x). \end{aligned}$$

If the last branch is entered while x remains an EFT coefficient, matching surface, effective operator display, or regime projection, then $\text{ValidHostileCertificate}_{SM}(x)$ fails.

Proof. Same-scope EFT claims that are coefficient, matching, operator-display, or regime-projection claims belong to T_{SM}^{reg} . If the claim is explicitly removed from T_{SM}^{reg} , it must be doing one of three things: bearing an unmet primitive burden, failing as primitive because the role is still coefficient/matching/projection status, or attempting a hostile certificate. A hostile certificate can close only if the object is no longer merely EFT display and instead supplies same-scope role necessity. At that point the claim is routed by the constructor partition through the relevant carrier sector, interface case, or target-update boundary. Thus EFT status itself cannot satisfy the validity predicate. $\square$

Corollary 9.5 (Wilson coefficient status). *A Wilson coefficient in the Standard Model-associated arena is a coefficient/projection status. Target-updating Wilsonian structure exits D_{SM}^3 , and hostile certificate attempts are eliminated by the no-valid-certificate theorem.*

Corollary 9.6 (Unification-plot status). *A unification plot or RG crossing is an RG-transport, scheme/matching, projection, burden, target-update, or hostile-certificate status. Crossing behavior does not become a primitive coupling carrier by graphical salience.*

Lemma 9.7 (Interface role-failure).

$$\begin{aligned} & \text{Interface}(x, C_i, C_j) \wedge \text{SameScope}_{D_{SM}^3}(x) \\ & \Rightarrow \text{SharedCarrier}(x) \vee x \in T_{SM}^{\text{reg}} \vee \text{CounterexampleBurden}_{SM}(x) \\ & \quad \vee \text{FailedPrimitive}_{SM}(x) \vee \text{HostileCertificateBranch}_{SM}(x). \end{aligned}$$

In addition,

$$\text{SharedCarrier}(x) \Rightarrow x \in C_{SM}^{\text{prim}} \vee x \downarrow C_{SM}^{\text{prim}}.$$

Proof. A same-scope interface either identifies a carrier role already shared across received packages, such as charge response; presents a translation, matching, or comparison status; bears an unmet burden; fails when the primitive claim collapses into mediation; or attempts hostile certification. Shared-carrier cases are already in C_{SM}^{prim} or descend directly to it. Registered interfaces contradict the nonregistered conjunct of a valid hostile certificate. Burden and failure do not establish primitive standing. A hostile interface attempt can close only if it is not merely an interface but an independently necessary carrier role, which is then handled by role-necessity exhaustion. $\square$

10 Numerical parameter and measurement closure

10.1 Numerical availability

A common objection identifies closure with derivation of all numerical values. This paper rejects that identification. Numerical values can be physically meaningful while remaining nonprimitive statuses.

Theorem 10.1 (Numerical availability does not confer primitive standing). *If x is a numerical Standard Model parameter, measured value, fit value, running value, pole value, phase coordinate, scheme coordinate, threshold value, or calibration value, then*

$$\text{NumericalStatus}(x) \wedge \neg \text{HostileCertificateBranch}_{SM}(x) \Rightarrow \neg \text{Prim}_{D_{SM}^3}(x).$$

Proof. Numerical status is status as value, coordinate, fit, measurement, calibration, or transport quantity. Each such role presupposes a carrier, route, scheme, or boundary condition relative to which the number is meaningful. Therefore numerical availability alone is downstream of standing rather than the source of primitive standing. A numerical object can challenge closure only by supplying an independent same-scope role certificate, not by being a number. $\square$

10.2 Masses and widths

Masses and widths are classified according to the carriers that license them. Gauge-boson masses are fixed relative to gauge/Higgs bridge structure. Quark masses and charged-lepton masses are matter/bridge response statuses. Neutrino mass parameters are neutral weak matter, mass-response, and flavor/oscillation boundary statuses. The Higgs pole mass is a Higgs bridge/fixation measurement status. None of these classifications derive the exact numerical value. They determine primitive status.

10.3 Couplings and running

The coupling values g_1, g_2, g_3, e, α , the weak mixing angle, running values, beta-function displays, matching thresholds, and scheme coordinates are CCR statuses. They descend to C_{CCR}^{prim} and registered transport. They are not primitive carriers merely because they are central to calculation or measurement.

10.4 Flavor numerical data

CKM and PMNS entries, mixing angles, CP phase coordinates, and neutrino ordering claims are transition coordinates, boundary reconstructions, phase coordinates, route statuses, or burden

branches. The primitive residue available to flavor is closed by $C_{\text{flavor}}^{\text{prim}}$, not by a numerical derivation of every entry.

10.5 Measurement bridge lemma

Lemma 10.2 (Measurement nonpromotion).

$$\text{MeasurementRecord}_{SM}(x) \Rightarrow x \in T_{SM}^{\text{reg}}.$$

Consequently a detector record, event excess, track, jet, bound, reconstruction, or empirical route does not become primitive by being evidentially salient.

Proof. Measurement records are indexed to apparatus, reconstruction route, calibration, statistical treatment, and carrier interpretation. They can confirm, pressure, or redirect a theory, but their status is evidential rather than primitive carrier standing. If a measurement is claimed to exceed evidential standing, the excess claim must be submitted as a hostile certificate attempt. A valid attempt would have to show role necessity; measurement salience alone cannot do that. $\square$

11 Higgs closure inside SM-6

11.1 Higgs primitive

SM-6 imports the Higgs endpoint carrier:

$$C_H = H_{br/fix}.$$

This carrier is the bridge/fixation standing role. It is not identical with scalar notation, vev notation, potential representation, pole mass, hierarchy comparison, metastability route, or extra-scalar model surface.

11.2 Higgs status atlas

Higgs-associated claim	SM-6 status	Primitive ruling
Scalar field notation	representative of bridge/fixation carrier	nonprimitive skin
Vacuum expectation value	fixation/scale presentation	nonprimitive status
Potential form	representative route and model display	nonprimitive status

Pole mass and width	measured scalar/bridge status	nonprimitive value
Hierarchy comparison	scale comparison or metric projection	nonprimitive comparison
Metastability	vacuum route status	nonprimitive route
Lower-vacuum branch	route, burden, or target-update status	no default primitive
Extra scalar	extension surface, target update, burden, or hostile branch	not automatic primitive
Composite/pNGB Higgs claim	representation, extension, target update, or hostile branch	not scalar-first primitive
Metric projection	comparison/projection status	nonprimitive interface

Theorem 11.1 (No fifth Higgs primitive). *For every same-scope Higgs-associated claim x ,*

$$\begin{aligned}
& HiggsClaim(x) \wedge SameScope_{D_{SM}^3}(x) \\
& \Rightarrow x \downarrow H_{br/fix} \vee x \in T_{SM}^{reg} \vee CounterexampleBurden_{SM}(x) \vee FailedPrimitive_{SM}(x) \\
& \quad \vee TargetUpdate_{SM}(x) \vee HostileCertificateBranch_{SM}(x).
\end{aligned}$$

No Higgs-associated attribute outside $H_{br/fix}$ retains primitive Higgs standing inside D_{SM}^3 . A contrary same-scope assertion must close a hostile certificate and thereby defeat SM-6.

Proof. The Higgs endpoint classifies same-scope Higgs-domain claims under bridge/fixation standing. Scalarity, vev, potential, scale, vacuum route, lower branch, extension surface, metric comparison, and identity redescription are representatives, routes, comparisons, dispositions, target updates, failures, or hostile submissions. A same-scope Higgs claim outside $H_{br/fix}$ either descends to the bridge/fixation carrier or lacks primitive standing. If it supplies independent primitive standing, it is the exposed hostile branch. $\square$

Theorem 11.2 (Higgs extension classification). *An extra scalar, role-splitting, composite, pNGB, multi-Higgs, or Higgs-sector BSM claim is classified as carrier-preserving enrichment, derivative representation, projection, extension surface, burden, failed primitive, target update, or hostile-certificate branch. It is not automatically a new primitive Higgs ground.*

Proof. Extension claims either preserve the bridge/fixation role while adding representatives or dynamics, replace the target by adding a new domain, fail to occupy the role they claim, or assert same-scope primitive standing outside $H_{br/fix}$. These are exactly the listed statuses. Automatic primitive promotion would bypass the bridge/fixation certificate and is therefore disallowed. $\square$

12 BSM and anomaly audit

12.1 BSM claim grammar

$\text{BSMClaim}(x)$ includes claims involving new particles, new symmetries, sterile neutrinos, fourth generations, dark-sector couplings, unification structures, supersymmetry-like structures, composite Higgs structures, extra dimensions, anomaly explanations, EFT operators, quantum-gravity interfaces, or other extension surfaces positioned against the Standard Model target.

SM-6 is not a model-by-model exclusion catalogue. It is a status audit. It asks what the BSM claim is doing relative to D_{SM}^3 : preserving carriers, projecting from them, updating the target, failing primitive standing, carrying a burden, or supplying a hostile certificate.

Theorem 12.1 (BSM target-update boundary).

$$\text{BSMClaim}(x) \wedge \neg \text{SameScope}_{D_{SM}^3}(x) \Rightarrow \text{TargetUpdate}_{SM}(x).$$

Proof. A BSM claim that changes the carrier domain, admissibility regime, primitive comparison class, or identity target does not remain inside D_{SM}^3 . It may be a correct extension or replacement of the Standard Model target, but relative to the SM-6 theorem it is a target update rather than a same-scope primitive survivor. $\square$

Theorem 12.2 (Same-scope BSM exhaustion). *For every same-scope BSM-associated claim x ,*

$$\begin{aligned} \text{BSMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \Rightarrow & \text{CarrierPreservingExtension}(x) \vee x \in T_{SM}^{\text{reg}} \\ & \vee \text{CounterexampleBurden}_{SM}(x) \vee \text{FailedPrimitive}_{SM}(x) \\ & \vee \text{HostileCertificateBranch}_{SM}(x). \end{aligned}$$

Proof. A same-scope BSM claim is evaluated inside the fixed SM carrier target. If it preserves the received carrier roles while adding dynamics, representation, interpretation, or model surface, it is carrier-preserving extension or registered status. If it supplies pressure without role closure, it is a burden. If its primitive assertion collapses into a representative, projection, route, parameter, or incoherent role, it is a failed primitive. If it asserts a same-scope primitive outside C_{SM}^{prim} , it is a hostile-certificate branch. Target-changing BSM claims were removed by the preceding theorem, so these cases exhaust same-scope BSM status. $\square$

Lemma 12.3 (Same-scope BSM certificate-failure). *Relative to the received sector closures, the Higgs endpoint, Official-Target Constructor Adequacy, and the target-update boundary,*

$$\text{BSMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \wedge x \notin C_{SM}^{\text{prim}} \Rightarrow \neg \text{ValidHostileCertificates}_{SM}(x).$$

Proof. The label “BSM” does not create an independent primitive route. A same-scope BSM claim must engage D_{SM}^3 through one or more constructors: gauge, matter, CCR, flavor, Higgs, generation governance, numerical/measurement status, projection/EFT status, anomaly surface, or interface. If it engages none of them, it is not same-scope and the target-update theorem applies. If it engages one of them, the relevant sector closure or SM-6 bridge lemma supplies carrier descent, registered status, governance, burden, or failure. Carrier-preserving extension gives descent or registered extension status; anomaly pressure gives evidence/burden status; projection and interface claims fail role necessity while they remain projection or interface functions. A same-scope BSM claim also cannot obtain primitive standing merely by claiming explanatory depth beneath the SM. If the proposed object is required for the same-scope SM role, it must coincide with, descend through, or reconstruct one of the certified carrier roles. If it does not, it has changed the primitive comparison class or the target. Thus a BSM hostile attempt lacks at least one validity conjunct inside D_{SM}^3 : nonregistered status, nonderivativeness, primitive standing, role necessity, fixed comparison class, or fixed same-scope membership. $\square$

Theorem 12.4 (Anomaly does not equal primitive promotion). *An anomaly is evidence pressure, repair demand, burden, target update, failed primitive, or hostile-certificate branch. It is not primitive standing merely by being anomalous.*

Proof. An anomaly is a discrepancy, tension, excess, or unexplained pattern relative to a measurement and model context. That status can be evidentially important, but it does not itself identify an irreducible carrier. If the anomaly is resolved within registered status, no primitive is added. If it changes the target, it is a target update. If it is claimed to demonstrate a same-scope primitive outside C_{SM}^{prim} , the hostile certificate burden applies. $\square$

Theorem 12.5 (Target update boundary).

$$\text{SMClaim}(x) \wedge \neg \text{SameScope}_{D_{SM}^3}(x) \Rightarrow \text{TargetUpdate}_{SM}(x).$$

Proof. Failure of same-scope status means the claim alters the domain, carrier class, admissibility standard, primitive comparison class, or identity target. Such a claim may be scientifically correct, but it does not occupy D_{SM}^3 . It is therefore a target update relative to SM-6. $\square$

Theorem 12.6 (Hostile certificate nonabsorption). *If x supplies a valid same-scope primitive role outside C_{SM}^{prim} , then the SM-6 deletion theorem is false.*

Proof. The deletion theorem asserts no same-scope primitive survivor outside C_{SM}^{prim} . A valid hostile certificate is exactly such a survivor. Therefore it cannot be absorbed by reclassification. It defeats the theorem. $\square$

13 Decision procedure and normal forms

13.1 Why a decision procedure is necessary

A global deletion theorem is vulnerable if its classifications are presented as after-the-fact labels. SM-6 therefore uses a decision procedure. The procedure does not decide whether future physics is true. It decides the status of a submitted claim relative to D_{SM}^3 . This distinction is essential. A claim may be empirically compelling and still be a target update. A claim may be useful in computation and still be a projection. A claim may be unresolved and still be a burden rather than a primitive survivor.

The decision procedure is also hostile-referee facing. It states where an adversary must press. The adversary may argue that the claim is same-scope, that it is primitive, that it is not derivative, that it is not registered, that it is not generation-governed, and that it does not change the target. If all of those tests close, the hostile branch closes and SM-6 fails. If one of them fails, the claim receives the corresponding nonprimitive or target-update status.

13.2 Precedence order

Definition 13.1 (SM-6 status precedence). For a submitted Standard Model-associated claim x , the status decision is evaluated in the following order:

$$\begin{aligned} &TargetUpdate \succ InvalidScope \succ HostileCertificate \succ CarrierDescent \\ &\succ RegisteredStatus \succ GenGovernance \succ Burden \succ FailedPrimitive. \end{aligned}$$

The order is not a hierarchy of importance. It is a typing discipline that prevents a claim from being counted in the wrong arena.

Target update is evaluated first because a claim outside D_{SM}^3 cannot be a survivor inside D_{SM}^3 . Hostile certification is evaluated before ordinary registration because a valid hostile primitive must not be buried in the registry. Carrier descent is evaluated before miscellaneous registration because a claim may be directly licensed by C_{SM}^{prim} . Generation governance is kept separate because it is neither primitive carrier status nor mere skin. Burdens and failures occupy the end of the procedure because they record unclosed or unsuccessful primitive submissions.

13.3 Decision procedure

Protocol 13.1 (SM-6 classification algorithm). Given x :

Step 1. Determine whether $SMClaim(x)$. If not, SM-6 does not classify x except as external to the claim arena.

- Step 2.** Determine whether $\text{SameScope}_{D_{SM}^3}(x)$. If not, assign $\text{TargetUpdate}_{SM}(x)$ relative to this theorem.
- Step 3.** Determine whether x attempts a same-scope primitive role outside C_{SM}^{prim} . If yes, enter $\text{HostileCertificateBranch}_{SM}(x)$ and evaluate $\text{ValidHostileCertificate}_{SM}(x)$. The no-valid-certificate theorem below shows that validity does not close inside D_{SM}^3 .
- Step 4.** Determine whether $x \downarrow C_{SM}^{\text{prim}}$. If yes, assign carrier descent and deny residual primitive standing.
- Step 5.** Determine whether $x \in T_{SM}^{\text{reg}}$. If yes, assign registered nonprimitive status.
- Step 6.** Determine whether $\text{GenGoverned}_3(x)$. If yes, assign governance status without primitive promotion.
- Step 7.** Determine whether x asserts primitive standing without a closed certificate. If yes, assign $\text{CounterexampleBurden}_{SM}(x)$.
- Step 8.** If the primitive assertion collapses into representative, route, coordinate, projection, target-update, or incoherent status, assign $\text{FailedPrimitive}_{SM}(x)$.

13.4 Normal form theorem

Theorem 13.1 (SM-6 normal form). *Every submitted same-scope Standard Model-associated claim has a normal form of the following kind:*

$$NF_{SM}(x) \in \{\text{CarrierDescent}, \text{RegisteredStatus}, \text{GenGovernance}, \\ \text{Burden}, \text{FailedPrimitive}, \text{HostileCertificate}\}.$$

Every submitted nonsame-scope Standard Model-associated claim has target-update normal form.

Proof. The claim partition lemma covers same-scope Standard Model-associated claims by sector, governance, value, measurement, projection, interface, BSM, anomaly, and overlap classes. Each class is assigned by a received theorem or an SM-6 bridge theorem to carrier descent, registered status, generation governance, burden, failure, or hostile branch. If same-scope fails, the target-update theorem applies. Thus every submitted claim has the stated normal form. $\square$

13.5 No hidden default primitive

The procedure has no default primitive branch. This is not a bias against discovery; it is the burden structure required by primitive standing. Primitive status is not inferred from novelty, unmodeled behavior, explanatory promise, model-building elegance, numerical surprise, or measurement significance. It is inferred from role occupancy. A claim that has not closed role occupancy cannot be counted as a survivor in a primitive deletion theorem.

Lemma 13.2 (Default nonpromotion). *A Standard Model-associated claim whose status is undetermined is not thereby a primitive survivor.*

Proof. Undetermined status is compatible with burden, evidence pressure, route insufficiency, or target-update uncertainty. Primitive survival is a positive status. It requires a certificate. Absence of final classification is not presence of primitive role occupancy. $\square$

14 Sector-by-sector global case atlas

14.1 Purpose of the atlas

The following atlas records how the global theorem handles the main claim families. It is not a replacement for the received sector proofs. It is a global audit map showing that no sector import is left floating and no obvious same-scope attribute class is unassigned.

14.2 Gauge-sector atlas

Gauge-associated claims include gauge-field notation, force-carrier particle labels, curvature displays, holonomy descriptions, transport routes, anomaly-compatibility claims, gauge-vacuum claims, weak-chiral labels, charge-response labels, gauge fixing, and BSM gauge surfaces. The primitive carriers available to this sector are the elements of C_{gauge} . Everything else is handled as representative, transport display, route, projection, anomaly status, gauge-vacuum status, burden, target update, or hostile certificate.

A photon label, for example, is not admitted as primitive because the word names an empirical and representational occupant. Its standing descends through gauge order, transport, curvature/holonomy, and charge-response roles. A gluon label similarly descends through color transport and gauge order. A weak-boson label descends through weak-chiral standing and the Higgs/bridge context needed for mass-related presentation. Gauge fixing is a coordinate or presentation discipline. A new gauge boson model is either an extension surface, target update, burden, or hostile certificate; it is not a primitive inside D_{SM}^3 by being new.

Proposition 14.1 (Gauge global routing). *Every same-scope gauge-associated claim outside C_{gauge} is routed to registered gauge status, carrier descent, BSM/anomaly burden, target update, failed primitive, or hostile branch.*

Proof. This is the SM-2 local deletion theorem transported through the sector embedding, with SM-6's BSM, anomaly, and target-update bridge rules added for global claims. Cross-sector claims involving matter, CCR, flavor, or Higgs structure are handled by overlap no-promotion. $\square$

14.3 Matter-sector atlas

Matter-associated claims include quark labels, charged-lepton labels, neutrino labels, spinor notation, representation assignments, color-bearing matter roles, neutral weak matter roles, mass-response slots, detector tracks, jets, hadron membership, oscillation routes, cosmological bounds, and generation labels. The primitive matter package is C_{matter} . Labels and tracks remain meaningful; they do not become primitive carriers.

A quark label descends to color-bearing matter and representation-response standing. A charged-lepton label descends to colorless charged matter. A neutrino label descends to neutral weak matter and, where relevant, flavor/oscillation routes. Jets and hadron membership are detector or bound-state route statuses. Cosmological bounds are projection/evidence statuses. Spinor notation is symbolic representative status. Mass labels occupy mass-response slots and Higgs/bridge realization; they are not exact numerical closure claims.

Proposition 14.2 (Matter global routing). *Every same-scope matter-associated claim outside C_{matter} is routed to matter carrier descent, detector/evidence status, route/projection status, generation governance, burden, failure, target update, or hostile branch.*

Proof. The SM-3 local deletion theorem closes primitive matter residue under D_{matter}^3 . The embedding into D_{SM}^3 preserves that status. Claims involving numerical masses, flavor transitions, Higgs fixation, or BSM matter extensions are handled by the relevant global bridge theorem and overlap no-promotion. $\square$

14.4 CCR atlas

CCR-associated claims include charge numbers, charge witnesses, charge-lattice claims, numerical coupling values, running couplings, beta functions, renormalization schemes, matching thresholds, calibration values, EW/EM bookkeeping, unification plots, fixed points, and EFT coefficients. The primitive CCR package is $C_{\text{CCR}}^{\text{prim}}$. Coupling values are not downgraded in physical importance; they are denied primitive carrier status by numerical availability alone.

The charge lattice and charge witness supply standing roles. A numerical value of g_i is a coupling modulus. A running curve is transport. A threshold is matching status. A scheme is a coordinate convention. An RG crossing is a projection or burden, not a primitive. A unification model may update the target or submit a hostile certificate; it does not become a primitive coupling carrier merely because the plotted curves meet.

Proposition 14.3 (CCR global routing). *Every same-scope CCR-associated claim outside $C_{\text{CCR}}^{\text{prim}}$ is routed to registered CCR status, projection/EFT status, carrier descent, burden, failed primitive, target update, or hostile branch.*

Proof. The SM-4 local closure proves that CCR attributes outside C_{CCR}^{prim} do not retain primitive status. SM-6 adds global handling for cross-sector measurements, EFT displays, unification plots, and BSM coupling surfaces. Numerical nonpromotion prevents measured or fitted values from becoming primitive carriers. $\square$

14.5 Flavor-sector atlas

Flavor-associated claims include CKM entries, PMNS entries, mixing angles, CP phase coordinates, rephasing conventions, basis choices, diagonalization, mass/flavor eigenstate labels, oscillation routes, FCNC claims, flavor anomalies, generation labels, and fourth-generation claims. The primitive flavor package is $C_{\text{flavor}}^{\text{prim}}$. The generation object remains governance.

A CKM entry is a transition coordinate. A PMNS entry is a boundary reconstruction or transition coordinate under neutrino-sector constraints. A raw phase coordinate is a skin; CP/holonomy residue is certified separately. Basis and rephasing choices are coordinate conventions. Oscillation routes are routes. A flavor anomaly is evidence pressure or burden. A fourth-generation assertion is a governance, extension, target-update, or hostile-certificate matter, not a free primitive.

Proposition 14.4 (Flavor global routing). *Every same-scope flavor-associated claim outside $C_{\text{flavor}}^{\text{prim}}$ is routed to transition-coordinate status, phase-coordinate status, route status, generation governance, burden, failure, target update, or hostile branch.*

Proof. The SM-5 local deletion theorem closes primitive flavor/mixing residue under fixed generation governance. The global generation-governance theorem keeps G_{gen}^3 outside C_{SM}^{prim} . BSM and anomaly claims enter the SM-6 audit matrix rather than becoming primitive by pressure. $\square$

14.6 Higgs-sector atlas

Higgs-associated claims include the scalar field, vev, potential, pole mass, width, Yukawa realization, hierarchy comparison, vacuum branch, metastability route, lower-vacuum claim, extra scalar, composite or pNGB presentation, role-splitting extension, and metric projection. The primitive Higgs carrier is $H_{\text{br}/\text{fix}}$.

The scalar field is a representative. The vev is a fixation presentation. The potential is a representative route. The pole mass and width are measured statuses. The hierarchy problem is a comparison and orientation problem, not a primitive scale carrier by itself. Metastability is a route status. A lower-vacuum claim is not a default primitive. Extra scalar and composite claims are extensions, target updates, burdens, failures, or hostile certificates.

Proposition 14.5 (Higgs global routing). *Every same-scope Higgs-associated claim outside $H_{\text{br}/\text{fix}}$ is routed to Higgs carrier descent, registered Higgs status, extension disposition, burden, failed primitive, target update, or hostile branch.*

Proof. The Higgs endpoint supplies the bridge/fixation carrier and no-residual-primitive-Higgs theorem. The extension classifier handles extra scalars, composite claims, and role-splitting claims. The scale and vacuum support papers prevent scale, hierarchy, metastability, and lower-branch descriptions from becoming primitive Higgs grounds by default. $\square$

15 Residual survivor audit

15.1 Candidate survivor form

A residual primitive survivor would need the form

$$S = (x, D_{SM}^3, \rho, \kappa),$$

where x is the submitted claim, ρ is the alleged primitive role, and κ is the certificate showing same-scope role occupancy outside C_{SM}^{prim} . Without κ , S is at most a burden. With κ , the submission must satisfy the full $\text{ValidHostileCertificates}_{SM}$ predicate. The no-valid-certificate theorem later proves that no such predicate closes inside the fixed constructor arena.

Lemma 15.1 (Survivor normal form). *Every alleged residual primitive survivor outside C_{SM}^{prim} must be representable as a hostile certificate submission.*

Proof. A survivor outside C_{SM}^{prim} must be primitive, same-scope, nonderivative, nonregistered, non-governance, and not a target update. These are exactly the core hostile-certificate conditions. Therefore an alleged survivor that is not a hostile certificate submission does not count as a survivor. $\square$

15.2 Audit deletion test

For a submitted x , the deletion test asks:

- (i) Does x belong to a received carrier package? If yes, it is not outside C_{SM}^{prim} .
- (ii) Does x descend to a received carrier package? If yes, it is derivative.
- (iii) Is x a registered status? If yes, no promotion follows.
- (iv) Is x generation-governed? If yes, governance separation applies.
- (v) Does x change the target? If yes, it is not same-scope.
- (vi) Does x attempt primitive standing without a certificate? If yes, it is a burden.
- (vii) Does x fail role occupancy? If yes, it is a failed primitive.
- (viii) Does x close a same-scope primitive role outside C_{SM}^{prim} ? If yes, it would be a valid hostile certificate; SM-6 proves this does not close inside D_{SM}^3 .

Theorem 15.2 (Residual survivor deletion test). *No candidate failing any of the first seven audit conditions can be counted as an element of*

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}).$$

A candidate apparently passing the eighth condition must satisfy ValidHostileCertificates_{SM}; inside the fixed constructor arena this is ruled out by the no-valid-certificate theorem. Outside that arena, the submission is either a target update or an external challenge to one of the dependencies.

Proof. The first condition removes elements of C_{SM}^{prim} from the complement. The second removes derivative descent. The third applies no registered-status promotion. The fourth applies governance no-leakage. The fifth applies target-update no-leakage. The sixth records burden rather than closure. The seventh denies primitive status. The eighth is not absorbed; it invokes the formal hostile-certificate predicate. Later sections prove that this predicate cannot be satisfied by any constructor class inside D_{SM}^3 . $\square$

16 Evidence tiers and burden discipline

16.1 Evidence is not primitive status

Empirical evidence is indispensable, but evidence and primitive standing are different statuses. Evidence may confirm a measurement, increase pressure on a model, motivate an extension, or force a target update. It does not automatically identify a primitive carrier. This is especially important for detector excesses, precision anomalies, flavor tensions, cosmological bounds, and fits of EFT coefficients.

Definition 16.1 (Evidence tiers). SM-6 distinguishes five evidence tiers:

- E1. registered measurement or calibration refinement;
- E2. anomaly or tension requiring explanation;
- E3. burden-bearing extension surface;
- E4. target-updating regime change;
- E5. hostile same-scope primitive certificate.

Only E5 threatens the deletion theorem directly.

Proposition 16.1 (Evidence-tier separation). *Tiers E1 through E4 do not by themselves establish $\text{Prim}_{D_{SM}^3}(x)$ for $x \notin C_{SM}^{\text{prim}}$. Tier E5, if valid, falsifies SM-6.*

Proof. E1 is measurement/calibration. E2 is pressure. E3 is burden. E4 changes target. None establishes same-scope primitive role occupancy outside C_{SM}^{prim} . E5 does establish the exact role, so the theorem is false if E5 closes. $\square$

16.2 Burden is a stable status

Burden status is not indecision. It is a formal status assigned to claims that have not discharged the primitive certificate. This allows SM-6 to remain open to new physics without surrendering primitive closure. A burden may later close as hostile certificate, fail, become registered, or become target update. Until then it is not a primitive survivor.

Lemma 16.2 (Burden preservation under future refinement). *Future refinement of a burden-bearing claim preserves burden status until the refinement closes one of the recognized branches: registered descent, failed primitive, target update, or hostile certificate.*

Proof. A burden records an unmet obligation. Additional data or modeling may discharge it, redirect it, or defeat it. But the obligation itself remains unmet until a recognized branch closes. Therefore burden cannot silently become primitive by accumulation of interest. $\square$

17 Global coverage theorem

17.1 Coverage statement

Theorem 17.1 (Global same-scope coverage). *For every same-scope Standard Model-associated claim x ,*

$$\begin{aligned} & \text{SMClaim}(x) \wedge \text{SameScope}_{D_{SM}^3}(x) \\ \Rightarrow & x \downarrow C_{SM}^{\text{prim}} \vee x \in T_{SM}^{\text{reg}} \vee \text{GenGoverned}_3(x) \\ & \vee \text{CounterexampleBurden}_{SM}(x) \vee \text{FailedPrimitive}_{SM}(x) \\ & \vee \text{HostileCertificateBranch}_{SM}(x). \end{aligned}$$

Proof. By the Attribute Membership Lock, the antecedent places x in $\text{Attr}(D_{SM}^3)$. By Constructor Exhaustion, x belongs to at least one of the constructor classes $\mathcal{A}_G, \mathcal{A}_M, \mathcal{A}_{CCR}, \mathcal{A}_F, \mathcal{A}_H, \mathcal{A}_{gen}, \mathcal{A}_{num}, \mathcal{A}_{meas}, \mathcal{A}_{\text{proj/EFT}}, \mathcal{A}_{\text{BSM/anom}}, \mathcal{A}_{int}, \mathcal{A}_{ov}$.

If x is generated by $\mathcal{A}_G$, the SM-2 import gives descent to C_{gauge} , registered gauge status, burden/failure status, or hostile branch. If generated by $\mathcal{A}_M$, the SM-3 import gives descent to C_{matter} , route/projection status, generation governance, burden/failure, or hostile branch. If generated by $\mathcal{A}_{CCR}$, the SM-4 import gives descent to C_{CCR}^{prim} or registered coupling/RG status. If generated by $\mathcal{A}_F$, the SM-5 import gives descent to $C_{\text{flavor}}^{\text{prim}}$, transition-coordinate status, generation governance, burden/failure, or hostile branch. If generated by $\mathcal{A}_H$, the Higgs endpoint gives descent to $H_{br/fix}$, registered Higgs status, burden/failure, target boundary, or hostile branch.

If generated by $\mathcal{A}_{gen}$, generation governance applies. If generated by $\mathcal{A}_{num}$, numerical nonpromotion assigns registered value, coordinate, calibration, fit, pole, or running status; hostile submissions

are evaluated separately. If generated by $\mathcal{A}_{meas}$, measurement nonpromotion assigns evidence, route, detector, or reconstruction status; hostile submissions are evaluated separately. If generated by $\mathcal{A}_{proj/EFT}$, projection role-failure and EFT role-failure assign registered projection, coefficient, matching, route status, burden, failure, or hostile branch. If generated by $\mathcal{A}_{BSM/anom}$, nonsame-scope submissions have already been removed by the target-update boundary; the remaining same-scope cases are carrier-preserving extension, registry, burden, failure, or hostile branch. If generated by $\mathcal{A}_{int}$, interface role-failure assigns shared carrier, registry, burden, failure, or hostile branch. If generated by $\mathcal{A}_{ov}$, overlap no-promotion applies. The stated same-scope disjunction follows without placing target updates inside the coverage theorem. $\square$

Lemma 17.2 (Constructor partition theorem). *Every same-scope Standard Model-associated claim falls into at least one constructor class in the presentation of $\text{Attr}(D_{SM}^3)$, and no constructor class lies outside the coverage theorem.*

Proof. The first part follows from Attribute Membership Lock plus Constructor Exhaustion. The second part follows by inspection of the coverage proof: each constructor is assigned to a received sector closure or to an SM-6 bridge lemma. Therefore no same-scope constructor remains outside the global coverage theorem. $\square$

17.2 Overlap discipline

A claim may fall into several classes. A measured CKM value is both flavor and numerical. A Higgs pole mass is both Higgs and numerical. An EFT coefficient may be both parameter-like and projection-like. A fourth-generation claim may be both matter-sector and BSM. Overlap does not generate primitive residue. It gives the claim multiple descents or registered statuses. A hostile certificate must be supplied separately and is evaluated by the validity predicate.

Lemma 17.3 (Overlap no-promotion). *If x has multiple nonprimitive registered descents, the conjunction of those descents does not promote x to primitive standing.*

Proof. Primitive standing is not the sum of many derivative statuses. Multiple dependence relations reinforce dependence. To become primitive, x must cease to be derivative in the relevant role and supply an independent carrier certificate. Without that certificate, overlap remains registered. $\square$

18 No-leakage theorem

18.1 Leakage definition

Definition 18.1 (Primitive leakage). $\text{Leak}_{SM}(x)$ holds when $x \notin C_{SM}^{\text{prim}}$ but x acquires $\text{Prim}_{D_{SM}^3}(x)$ without satisfying the hostile-certificate branch.

Theorem 18.1 (No registered-to-primitive leakage).

$$x \in T_{SM}^{\text{reg}} \Rightarrow \neg \text{Leak}_{SM}(x).$$

Proof. A registered status is assigned only to nonprimitive roles: representative, coordinate, value, route, projection, calibration, scheme, evidence status, extension surface, or anomaly pressure. Such a status cannot leak into primitive standing without changing its status assignment. The only permitted route for such a challenge is a hostile certificate. Leakage by silent promotion is disallowed. $\square$

Theorem 18.2 (Governance no-leakage).

$$\text{GenGoverned}_3(x) \Rightarrow \neg \text{Leak}_{SM}(x).$$

Proof. Generation governance fixes how generation-indexed claims are handled inside D_{SM}^3 . Governance is not a carrier. Therefore governance cannot leak into primitive status. A claim that asserts more than governance must either update the target or supply a hostile certificate. $\square$

Theorem 18.3 (Burden and failure no-leakage).

$$\text{CounterexampleBurden}_{SM}(x) \Rightarrow \neg \text{Prim}_{D_{SM}^3}(x)$$

until the burden is discharged by a valid hostile certificate, and

$$\text{FailedPrimitive}_{SM}(x) \Rightarrow \neg \text{Prim}_{D_{SM}^3}(x).$$

Proof. A burden is not a closed certificate. It marks the obligation to demonstrate primitive role occupancy. Until that obligation closes, primitive status is not established. A failed primitive attempt is explicitly a failure to establish primitive status. Neither leaks into primitive standing. $\square$

Theorem 18.4 (Target-update no-leakage).

$$\text{TargetUpdate}_{SM}(x) \Rightarrow \neg \text{SameScope}_{D_{SM}^3}(x).$$

Consequently target updates cannot be residual same-scope primitive escapes inside D_{SM}^3 .

Proof. Target updates are defined by changing the target domain, carrier class, admissibility standard, or identity object. Such claims may replace or extend the Standard Model target, but they do not remain inside D_{SM}^3 . They therefore cannot be survivors in $\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}$. $\square$

19 Elimination of hostile certificate validity

The hostile branch is essential for falsifiability, but the deletion theorem also requires that no hostile certificate close inside the fixed constructor arena. This section gives the proof in constructor-local form. Role-necessity exhaustion is derived from local sector and bridge lemmas; it is not used as an unexplained restatement of the endpoint.

19.1 Role necessity and local exhaustion

Definition 19.1 (Role necessity). $\text{RoleNecessity}_{SM}(x)$ holds when the alleged role of x is necessary for same-scope Standard Model carrier standing in D_{SM}^3 , not merely useful for calculation, measurement, display, matching, compression, explanation, or model-building.

For constructor-local statements, write $\text{RoleNecessity}_i(x)$ for $\text{RoleNecessity}_{SM}(x) \wedge x \in \mathcal{A}_i$.

Lemma 19.1 (Gauge role-necessity exhaustion).

$$\text{RoleNecessity}_G(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x \in C_{\text{gauge}} \vee x \downarrow C_{\text{gauge}}.$$

Proof. The gauge-sector closure imports the complete same-scope gauge primitive package. Gauge-field labels, curvature displays, holonomy descriptions, transport presentations, anomaly reports, and gauge-vacuum descriptions either occupy one of the certified gauge roles or descend to them. A gauge object outside the package is a representation, route, projection, registry entry, burden, failure, or target update, not a further role necessity. $\square$

Lemma 19.2 (Matter role-necessity exhaustion).

$$\text{RoleNecessity}_M(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x \in C_{\text{matter}} \vee x \downarrow C_{\text{matter}}.$$

Proof. The matter-sector closure imports the matter representation, color-bearing, colorless charged, neutral weak, and mass-response-slot roles under fixed generation governance. Tracks, jets, hadron membership, mass labels, oscillation routes, and detector projections are licensed by those roles or by registered status. They do not add a further matter-role necessity. $\square$

Lemma 19.3 (CCR role-necessity exhaustion).

$$\text{RoleNecessity}_{CCR}(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x \in C_{CCR}^{\text{prim}} \vee x \downarrow C_{CCR}^{\text{prim}}.$$

Proof. The CCR closure imports charge-response, charge-witness, and charge-lattice standing as the primitive CCR package. Coupling moduli, running values, RG curves, scheme coordinates, matching thresholds, calibration values, beta-function displays, and unification plots are transport

or coordinate statuses. If any of them is claimed as necessary for same-scope standing, the necessity is the received CCR carrier role or descends to it. $\square$

Lemma 19.4 (Flavor role-necessity exhaustion).

$$\text{RoleNecessity}_F(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x \in C_{\text{flavor}}^{\text{prim}} \vee x \downarrow C_{\text{flavor}}^{\text{prim}}.$$

Proof. The flavor-sector closure imports flavor-transition standing and CP/holonomy-residue standing. CKM and PMNS entries, raw phase coordinates, basis conventions, rephasing choices, diagonalization routes, oscillation routes, FCNC surfaces, and flavor anomalies are coordinates, routes, projections, burdens, failures, or target updates. Same-scope flavor role necessity is therefore exhausted by membership in or descent to $C_{\text{flavor}}^{\text{prim}}$. $\square$

Lemma 19.5 (Higgs role-necessity exhaustion).

$$\text{RoleNecessity}_H(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x = H_{br/fix} \vee x \downarrow H_{br/fix}.$$

Proof. The Higgs endpoint imports $H_{br/fix}$ as bridge/fixation standing. Scalar notation, vev notation, potential form, pole mass, hierarchy comparison, vacuum branch, metastability route, lower-vacuum branch, extra-scalar surface, and metric projection are representatives, routes, comparisons, extension dispositions, burdens, failures, or target updates. A same-scope Higgs role necessity is therefore $H_{br/fix}$ itself or descends to it. $\square$

Lemma 19.6 (Generation, value, and measurement role-failure). *If $x \in \mathcal{A}_{gen} \cup \mathcal{A}_{num} \cup \mathcal{A}_{meas}$, $x \in \text{Attr}(D_{SM}^3)$, and $\neg \text{TargetUpdate}_{SM}(x)$, then any asserted carrier role necessity is either routed through the matter/flavor/Higgs/CCR carrier cases or fails as governance, value, or evidence status.*

Proof. Generation claims govern family indexing; they do not supply an additional primitive carrier. Numerical claims are values, fits, coordinates, pole values, running values, calibration values, or boundary reconstructions. Measurement claims are evidence, detector, route, reconstruction, bound, or excess statuses. If a claim in this group asserts more than governance, value, or evidence, that excess is not produced by the group itself and must be evaluated by the relevant sector, projection/EFT, BSM/anomaly, interface, or target-update constructor. $\square$

Lemma 19.7 (Projection/EFT local role-failure). *If $x \in \mathcal{A}_{\text{proj/EFT}}$, $x \in \text{Attr}(D_{SM}^3)$, and $\neg \text{TargetUpdate}_{SM}(x)$, then x supplies projection, matching, coefficient, effective-operator, display, or route status. Such status does not satisfy $\text{RoleNecessity}_{SM}(x)$. If role necessity is asserted independently, the claim is routed through a certified carrier constructor or fails the fixed comparison class.*

Proof. Projection and EFT constructions mediate, display, compare, match, or parametrize standing supplied elsewhere. Those functions can be indispensable for calculation without being primitive

carrier roles. A projection/EFT object that is truly required for the same-scope carrier role either identifies a certified carrier already in C_{SM}^{prim} , descends to it, or changes the target/comparison class. Remaining projection/EFT objects are registered, burdened, or failed. $\square$

Lemma 19.8 (BSM/anomaly local role-failure). *If $x \in \mathcal{A}_{\text{BSM/anom}}$, $x \in \text{Attr}(D_{SM}^3)$, and $\neg \text{TargetUpdate}_{SM}(x)$, then a same-scope role-necessity assertion for x is routed through a sector, Higgs, generation, numerical, measurement, projection/EFT, interface, or overlap constructor. The BSM/anomaly label itself supplies no primitive role necessity.*

Proof. A same-scope anomaly is evidence pressure, repair demand, burden, failed primitive, or hostile attempt. A same-scope BSM surface preserves carriers, registers a model/projection/status, bears a burden, fails, or attempts a certificate through an actual constructor. Explanatory depth beneath the SM is not primitive occupation by itself. If the proposed object is required for an SM role, it must coincide with, descend through, or reconstruct one of the certified carrier roles; otherwise it changes the primitive comparison class or target. $\square$

Lemma 19.9 (Interface and overlap local role-failure). *If $x \in \mathcal{A}_{\text{int}} \cup \mathcal{A}_{\text{ov}}$, $x \in \text{Attr}(D_{SM}^3)$, and $\neg \text{TargetUpdate}_{SM}(x)$, then a same-scope role-necessity assertion for x either identifies a shared carrier in or descending to C_{SM}^{prim} , or fails as mediation, translation, comparison, or overlap aggregation.*

Proof. An interface mediates between carrier roles; overlap records multiple constructor memberships. Neither mediation nor overlap multiplication creates irreducible carrier standing. Shared-carrier cases are already in or descend to C_{SM}^{prim} . Nonshared cases are registered, burdened, or failed. A proposed interface primitive that changes the comparison class is a target update rather than a same-scope survivor. $\square$

Theorem 19.10 (Role-Necessity Exhaustion inside D_{SM}^3).

$$\text{RoleNecessity}_{SM}(x) \wedge x \in \text{Attr}(D_{SM}^3) \wedge \neg \text{TargetUpdate}_{SM}(x) \Rightarrow x \in C_{SM}^{\text{prim}} \vee x \downarrow C_{SM}^{\text{prim}}.$$

Proof. By Official-Target Constructor Adequacy, x belongs to at least one constructor class. In the gauge, matter, CCR, flavor, and Higgs cases, the constructor-local role-necessity lemmas place x in or descending to the corresponding sector carrier package, hence in or descending to C_{SM}^{prim} . In the generation, numerical, measurement, projection/EFT, BSM/anomaly, interface, and overlap cases, the local role-failure lemmas show that the constructor either does not supply carrier role necessity or routes the asserted necessity through a certified carrier constructor. The theorem assumes role necessity and excludes target update. Therefore only the certified-carrier or carrier-descent alternatives remain. $\square$

19.2 Certificate-conjunct failure table

A valid hostile certificate requires all conjuncts in the valid-hostile-certificate definition. The following table records which conjunct fails for each constructor class when $x \notin C_{SM}^{\text{prim}}$.

Constructor	Ruling inside D_{SM}^3	Failed certificate conjunct
$\mathcal{A}_G$	gauge residue closed by SM-2	primitive standing or non-descent
$\mathcal{A}_M$	matter residue closed by SM-3	primitive standing or non-descent
$\mathcal{A}_{CCR}$	couplings/RG/schemes enter CCR	nonregistered status or
	descent or registry	non-descent
$\mathcal{A}_F$	flavor coordinates/routes governed by SM-5	primitive standing, nonregistered status, or nongovernance
$\mathcal{A}_H$	Higgs claims descend to $H_{br/fix}$ or registry	primitive standing or non-descent
$\mathcal{A}_{gen}$	family-index claims are governed	$\neg \text{GenGoverned}_3(x)$
$\mathcal{A}_{num}$	values/fits/running quantities are registered	$x \notin T_{SM}^{\text{reg}}$
$\mathcal{A}_{meas}$	events, tracks, excesses, and bounds are evidence statuses	$x \notin T_{SM}^{\text{reg}}$ or primitive standing
$\mathcal{A}_{\text{proj/EFT}}$	projection/EFT objects are registered, burdened, failed, or routed to certified carriers	role necessity, nonregistered status, or non-descent
$\mathcal{A}_{\text{BSM/anom}}$	same-scope BSM/anomaly claims reduce to constructors, registry, burden, failure, or certified-carrier descent	role necessity, primitive standing, nonregistered status, or fixed comparison class
$\mathcal{A}_{int}$	interfaces are shared carriers or registered mediation	non-descent, nonregistered status, or role necessity
$\mathcal{A}_{ov}$	overlap multiplies dependence relations, not primitive standing	primitive standing or role necessity

Lemma 19.11 (Sector constructor certificate failure). *If $x \in \mathcal{A}_G \cup \mathcal{A}_M \cup \mathcal{A}_{CCR} \cup \mathcal{A}_F \cup \mathcal{A}_H$, $x \in \text{Attr}(D_{SM}^3)$, and $x \notin C_{SM}^{\text{prim}}$, then $\neg \text{ValidHostileCertificate}_{SM}(x)$.*

Proof. The received sector deletion theorems and the Higgs endpoint assign noncarrier sector attributes to carrier descent, registered status, governance, burden, failure, or target boundary. Thus at least one hostile-certificate conjunct fails: $\text{Prim}_{D_{SM}^3}(x)$, $\neg(x \downarrow C_{SM}^{\text{prim}})$, $x \notin T_{SM}^{\text{reg}}$, or $\neg \text{GenGoverned}_3(x)$. The global sector-embedding theorem preserves these local rulings in D_{SM}^3 . $\square$

Lemma 19.12 (Registered and governed constructor certificate failure). *If $x \in \mathcal{A}_{gen} \cup \mathcal{A}_{num} \cup \mathcal{A}_{meas}$*

and $x \in \text{Attr}(D_{SM}^3)$, then $\neg \text{ValidHostileCertificate}_{SM}(x)$.

Proof. Generation constructor claims satisfy $\text{GenGoverned}_3(x)$, with the value three supplied by the open-generation endpoint, contradicting $\neg \text{GenGoverned}_3(x)$. Numerical constructor claims are registered as values, fits, coordinates, pole values, running values, or calibration values, contradicting $x \notin T_{SM}^{\text{reg}}$. Measurement constructor claims are registered as evidence, detector, route, reconstruction, bound, or excess statuses. If a submitted object is no longer governed, numerical, or evidential, it is not covered by this constructor typing and is evaluated by the constructor that supplies the asserted excess role. The governed, numerical, or evidential status itself cannot close hostile validity. $\square$

Lemma 19.13 (Projection/EFT constructor certificate failure). *If $x \in \mathcal{A}_{\text{proj/EFT}}$, $x \in \text{Attr}(D_{SM}^3)$, and $x \notin C_{SM}^{\text{prim}}$, then $\neg \text{ValidHostileCertificate}_{SM}(x)$.*

Proof. Projection role-failure and EFT role-failure show that same-scope projection/EFT claims are registered, burdened, failed, or hostile attempts. Registered cases fail $x \notin T_{SM}^{\text{reg}}$. Burden cases fail $\text{Prim}_{D_{SM}^3}(x)$ because no certificate has closed. Failure cases deny primitive status. Hostile attempts fail $\text{RoleNecessity}_{SM}(x)$ while they remain projection, coefficient, matching, operator-display, or route statuses. If role necessity is asserted independently, Role-Necessity Exhaustion forces membership in or descent to C_{SM}^{prim} , contradicting $x \notin C_{SM}^{\text{prim}} \wedge \neg(x \downarrow C_{SM}^{\text{prim}})$, or the claim leaves D_{SM}^3 as a target update. $\square$

Lemma 19.14 (BSM/anomaly constructor certificate failure). *If $x \in \mathcal{A}_{\text{BSM/anom}}$, $x \in \text{Attr}(D_{SM}^3)$, and $x \notin C_{SM}^{\text{prim}}$, then $\neg \text{ValidHostileCertificate}_{SM}(x)$.*

Proof. Because $x \in \text{Attr}(D_{SM}^3)$, nonsame-scope BSM claims have already been removed by the target-update boundary. The same-scope BSM certificate-failure lemma applies: a BSM or anomaly label must engage the carrier architecture through the sector, Higgs, generation, numerical, measurement, projection/EFT, interface, or overlap constructors. The relevant constructor ruling supplies carrier descent, registered status, governance, burden, or failure. If the BSM/anomaly claim asserts independent role necessity, Role-Necessity Exhaustion sends it to C_{SM}^{prim} or descent to C_{SM}^{prim} , contradicting the hostile-certificate complement and non-descent conjuncts. Fixed-comparison failure routes to target update or failed primitive status. Hence no BSM/anomaly constructor claim satisfies all validity conditions. $\square$

Lemma 19.15 (Interface and overlap constructor certificate failure). *If $x \in \mathcal{A}_{\text{int}} \cup \mathcal{A}_{\text{ov}}$, $x \in \text{Attr}(D_{SM}^3)$, and $x \notin C_{SM}^{\text{prim}}$, then $\neg \text{ValidHostileCertificate}_{SM}(x)$.*

Proof. Interface role-failure sends interface claims to shared carrier, registered mediation, burden, failure, or hostile attempt. Shared-carrier cases are in or descend to C_{SM}^{prim} ; registered mediation fails $x \notin T_{SM}^{\text{reg}}$; burden and failure do not establish primitive standing; and hostile interface attempts fail role necessity while they remain mediation. If they identify a necessary carrier,

Role-Necessity Exhaustion sends them to C_{SM}^{prim} or descent. Overlap cases inherit the strongest applicable constructor ruling. Multiple descents, values, projections, and measurement statuses multiply dependence relations rather than create a new irreducible role, so primitive standing or role necessity fails; a submission that states an additional carrier role is evaluated through the already handled carrier or target-update paths. $\square$

Theorem 19.16 (No Valid Same-Scope Hostile Certificate).

$$\neg \exists x \text{ ValidHostileCertificate}_{SM}(x).$$

Equivalently,

$$\forall x (x \in \text{Attr}(D_{SM}^3) \wedge x \notin C_{SM}^{\text{prim}} \Rightarrow \neg \text{ValidHostileCertificate}_{SM}(x)).$$

Proof. Assume $x \in \text{Attr}(D_{SM}^3)$ and $x \notin C_{SM}^{\text{prim}}$. By Official-Target Constructor Adequacy, x belongs to at least one constructor class. If x belongs to a sector constructor, Sector Constructor Certificate Failure applies. If it belongs to a governed, numerical, or measurement constructor, Registered and Governed Constructor Certificate Failure applies. If it belongs to the projection/EFT constructor, Projection/EFT Constructor Certificate Failure applies. If it belongs to the BSM/anomaly constructor, BSM/Anomaly Constructor Certificate Failure applies. If it belongs to the interface or overlap constructor, Interface and Overlap Constructor Certificate Failure applies. Each case blocks at least one conjunct of $\text{ValidHostileCertificate}_{SM}(x)$. Therefore no x satisfies the validity predicate inside D_{SM}^3 . $\square$

Corollary 19.17 (Hostile branch closure inside the fixed arena).

$$\text{HostileCertificateBranch}_{SM}(x) \wedge x \in \text{Attr}(D_{SM}^3) \Rightarrow \neg \text{ValidHostileCertificate}_{SM}(x).$$

Proof. A hostile branch is a certificate attempt. The preceding theorem proves that no such attempt satisfies the validity predicate inside the constructor-presented arena. A claimed counterexample must therefore break an imported dependency, break constructor adequacy, or update the target rather than survive as an internal same-scope primitive. $\square$

20 Final residual deletion theorem

Theorem 20.1 (Standard Model Carrier Exhaustion Closure).

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset.$$

Equivalently, the residual same-scope primitive survivor count is zero.

Proof. Assume for contradiction that there exists

$$y \in \text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}$$

such that $\text{Prim}_{D_{SM}^3}(y)$. By the Attribute Membership Lock, y is a same-scope Standard Model-associated attribute. By Constructor Exhaustion, y belongs to at least one constructor class. By Global Same-Scope Coverage, y is assigned to one of the following statuses:

$$y \downarrow C_{SM}^{\text{prim}} \vee y \in T_{SM}^{\text{reg}} \vee \text{GenGoverned}_3(y) \vee \text{CounterexampleBurden}_{SM}(y) \vee \text{FailedPrimitive}_{SM}(y) \\ \vee \text{HostileCertificateBranch}_{SM}(y).$$

If $y \downarrow C_{SM}^{\text{prim}}$, then y is derivative of the carrier package and cannot be a residual primitive outside it. If $y \in T_{SM}^{\text{reg}}$, no registered-status promotion applies. If $\text{GenGoverned}_3(y)$, governance no-leakage applies. If $\text{CounterexampleBurden}_{SM}(y)$, the burden has not established primitive standing. If $\text{FailedPrimitive}_{SM}(y)$, primitive standing is denied.

The only remaining case is $\text{HostileCertificateBranch}_{SM}(y)$. By Survivor Normal Form, a residual primitive survivor outside C_{SM}^{prim} must close the hostile-certificate predicate. If the branch is invalid, it does not establish $\text{Prim}_{D_{SM}^3}(y)$. If it is valid, then $\text{ValidHostileCertificate}_{SM}(y)$, contradicting the No Valid Same-Scope Hostile Certificate theorem. Role-Necessity Exhaustion also blocks validity directly: any necessary same-scope role is in or descends to C_{SM}^{prim} , contrary to residual nonmembership and non-descent. Therefore the hostile branch yields no survivor inside D_{SM}^3 .

Every case contradicts residual primitive survival for y . Hence no such y exists, and

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset.$$

□

Corollary 20.2 (Primitive closure, skin openness). *The Standard Model carrier arena is primitively closed at the same-scope carrier level while remaining open at the level of measurement, numerical fitting, projection, EFT display, anomaly pressure, BSM audit, and target update.*

21 Prediction, falsification, and evidence register

21.1 Prediction and falsification as target-preserving contact

Prediction and falsification are not merely internal bookkeeping statuses. They presuppose preserved target identity, reference-bearing standing, and comparison class. A prediction is a prediction of the same claim only under sameness preservation; a falsification is a falsification of the same claim only under target-preserving failure.

Theorem 21.1 (Prediction and falsification require non-degenerate reference).

$$\text{Prediction}_{SM}(x) \vee \text{Falsification}_{SM}(x) \Rightarrow \text{KernelPreserved}(x).$$

More explicitly,

$$\text{Prediction}_{SM}(x) \Rightarrow \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x) \wedge \text{ReferenceStanding}(x),$$

while

$$\text{Falsification}_{SM}(x) \Rightarrow \text{SameClaimUnderTest}(x) \wedge \text{FailureAgainstPreservedTarget}(x).$$

Proof. A prediction without a preserved target is only a future description. A falsification without a preserved claim under test is only a mismatch after target replacement. Prediction and falsification therefore require the kernel conditions that preserve standing, reference, sameness, comparison class, and failure against the same object. $\square$

Corollary 21.2 (Reality-contact discipline). *The prediction register in SM-6 is not a detached taxonomy of future sentences. It is an audit of how future theory-world confrontations preserve, burden, fail, or target-update the same Standard Model carrier claim. If the target is not preserved, the result is not a same-scope falsification; it is a target update or an attack on the dependency package.*

21.2 Structural predictions

SM-6 makes structural predictions about status, not numerical predictions about future measured values. These predictions have reality contact because they are claims about target-preserving future confrontations: new measurements, anomaly reports, EFT deviations, and BSM surfaces must either preserve the same carrier target, burden it, fail as primitives, or update it. A valid same-scope primitive certificate would defeat the theorem rather than be absorbed.

Future claim type	Expected SM-6 status	What would falsify SM-6
New Higgs scalar	extension surface, target update, burden, or hostile branch	same-scope nonderivative primitive Higgs ground outside $H_{br/fix}$
Flavor anomaly	evidence pressure, transition-coordinate issue, burden, or target update	new primitive transition carrier outside $C_{\text{flavor}}^{\text{prim}}$

Coupling convergence	RG/projection/scheme status or burden	primitive coupling carrier outside C_{CCR}^{prim}
Fourth generation	governance branch, extension, target update, or hostile branch	same-scope primitive generation carrier
EFT deviation	coefficient, projection, or matching or target update	same-scope primitive EFT carrier outside C_{SM}^{prim}
Detector excess	evidence, anomaly, burden status	direct role-occupancy primitive certificate
New symmetry	extension, target update, or hostile branch	same-scope primitive outside C_{SM}^{prim}
Neutrino anomaly	neutral-weak/flavor route status, burden, or target update	primitive neutral matter or transition role outside package
Dark-sector coupling	extension surface, projection, target update, or hostile branch	same-scope nonderivative primitive carrier inside D_{SM}^3

21.3 Falsification protocol

Protocol 21.1 (Hostile falsification). A falsifying case must establish all of the following:

- (1) same-scope membership inside D_{SM}^3 ;
- (2) primitive standing;
- (3) nonderivativeness from C_{SM}^{prim} ;
- (4) nonmembership in T_{SM}^{reg} ;
- (5) nongovernance by G_{gen}^3 ;
- (6) non-target-update status;
- (7) role-occupancy necessity;
- (8) compatibility with the fixed comparison class.

This protocol is intentionally stringent. It is not enough to name an unfamiliar particle, value, anomaly, operator, or model surface. The case must show why the claim is a same-scope primitive carrier inside the fixed Standard Model domain rather than a registered object, extension, burden, failure, or target update.

22 Hostile reconstruction

Proposition 22.1 (Exhaustive failure modes). *A successful objection to SM-6 must do at least one of the following: break an imported sector deletion theorem; break the Higgs endpoint; break the*

standalone structural-spine embedding; break the constructor presentation of $\text{Attr}(D_{SM}^3)$; show that a registered status can promote to primitive without a valid certificate; produce a valid same-scope hostile certificate; show that a target update has been misclassified as same-scope; or show that a same-scope claim has been incorrectly expelled as target update.

Proof. The final theorem depends exactly on the received deletions, the Higgs import, the structural embedding, the constructor partition, the nonpromotion lemmas, the no-valid-certificate theorem, and the target-update boundary. Breaking none of these leaves the proof path intact. Breaking one of them identifies the corresponding load-bearing failure mode. $\square$

Proposition 22.2 (Kernel denial burden).

$$\begin{aligned} \text{DenyKernel}(x) \Rightarrow & \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Proof. The kernel supplies the minimal non-degeneracy conditions for same-claim preservation, target identity, reference-bearing comparison, and target-preserving failure. A denial of the kernel is therefore not a simple framework rejection. It must recover the functions the kernel supplies: sameness without collapse into redescription, standing without arbitrary assertion, reference without target drift, and prediction/falsification without equivocation between failure and replacement. $\square$

Remark 22.1 (Falsifiability and no-valid-certificate closure). Within the received dependency package and constructor-presented arena, SM-6 proves that no valid hostile certificate closes. A future submission that appears to close such a certificate is not classified away. It attacks at least one load-bearing item: an imported sector theorem, the Higgs endpoint, constructor adequacy, role-necessity exhaustion, target/same-scope classification, or an SM-6 bridge lemma. If that attack succeeds, SM-6 is false.

22.1 Classifier-immunization objection

The strongest hostile objection says that SM-6 classifies away every new primitive by calling it a skin, burden, or target update. The response is the hostile-certificate branch. A valid same-scope primitive outside C_{SM}^{prim} defeats the theorem. The classifier rejects primitive promotion only when the submitted claim lacks role-occupancy standing, is derivative, is registered, is governed, fails, or changes the target.

22.2 Dependency objection

A second objection says that SM-6 depends on prior AASC sector papers. It does. The proof is dependency-locked. This is not a defect. The valid attack is to break an imported theorem, break a

domain embedding, break a bridge lemma, or supply a faithful hostile certificate. Requiring SM-6 to re-prove every local paper would confuse synthesis with repetition.

22.3 No-new-physics objection

A third objection says primitive closure implies no new physics. It does not. New physics may update the target, add sectors, introduce new measurements, alter anomalies, or demand a new comparison class. SM-6 says only that such physics is not an unlicensed same-scope primitive inserted inside D_{SM}^3 . A valid hostile certificate would defeat the theorem rather than be absorbed by it.

22.4 Parameter objection

A fourth objection says the Standard Model has undetermined parameters and therefore cannot be closed. This equivocates between numerical derivation and primitive-status closure. SM-6 does not derive the parameter table. It classifies numerical parameters as moduli, coordinates, calibration values, running values, boundary reconstructions, or measurement statuses.

22.5 Generation objection

Generation cardinality is received through Proposition 3.1. Its input architecture is classified by original-work necessity before counting, and its complete occupation is constructed from the shared packet. Rank one follows from that architecture. The governing count is therefore a received theorem with explicit duty, incidence, and occupation hypotheses, rather than an empirical family-count input. An additional primitive duty would require its own joint receipt; an enlarged physical inventory does not by itself supply that duty or add a generation carrier to C_{SM}^{prim} .

22.6 Higgs objection

A sixth objection says the scalar field, vev, and potential are primitives. SM-6 rejects that promotion. They are representatives and route/status displays licensed by $H_{br/fix}$. Their physical use is preserved; primitive standing is denied.

22.7 Projection-interface objection

A seventh objection says the synthesis requires a projection primitive. SM-6 denies promotion by mediation. Projection, comparison, matching, display, and measurement functions are registered interface statuses. A submission asserting independent same-scope primitive role occupancy must pass the hostile-certificate predicate, which the fixed-domain theorem eliminates internally.

22.8 Hidden-BSM-primitive objection

An eighth objection says a BSM proposal may preserve the SM target while revealing a deeper same-scope primitive beneath the certified carrier architecture. Explanatory depth is not primitive occupation. If the proposed object is required for the same-scope SM role, it must coincide with, reconstruct, or descend through one of the certified carriers. If it does not, it changes the target or primitive comparison class, remains a burdened hostile attempt, or fails certificate validity.

23 Scope, nonclaims, and final interpretation

23.1 What is closed

Closed are same-scope primitive residue, skin-to-primitive promotion, unlicensed parameter primitive status, unlicensed projection primitive status, unlicensed BSM primitive insertion, unlicensed Higgs primitive escape, unlicensed flavor/generation primitive escape, and unlicensed cross-sector interface primitive standing.

23.2 What remains open

Open are empirical refinement, new measurements, anomaly reports, precision fits, EFT coefficients, BSM target updates, new regimes, numerical derivations, parameter values, and physics outside D_{SM}^3 . Same-scope generation cardinality is closed by the open-generation endpoint; target-updating family structures remain open. The theorem preserves this openness by sorting it rather than denying it.

23.3 Final interpretation

The Standard Model can be complete in one sense and open in another. It is closed as a same-scope primitive carrier architecture. It remains open as empirical, numerical, projective, and target-updating physics. Future physics may change measurements, add projections, generate anomalies, introduce BSM surfaces, or update the target. What it cannot do, without defeating the theorem, is add an unlicensed same-scope primitive inside the closed carrier architecture.

24 Conclusion

SM-6 completes the carrier-exhaustion arc. It receives the classifier, sector closures, open-generation cardinality endpoint, Higgs endpoint, and structural spine; builds the global carrier package; registers

nonprimitive statuses; separates generation governance; audits projection, EFT, numerical, Higgs, BSM, and anomaly escape routes; exposes hostile certificates; and proves residual deletion. The endpoint is

$$\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset.$$

The result is not empirical finality. It is primitive closure with skin openness. The kernel used by the proof is not an optional overlay on physics; it is the minimal necessity condition under which sameness, standing, reference, comparison, prediction, and falsification remain non-degenerate. Under those conditions, the Standard Model carrier domain is exhausted at the same-scope primitive level, while measurement, projection, anomaly, EFT, BSM, and target-updating physics remain open.

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A Lean 4 formalization target and public audit reference

The SM ARC archive [R14] is a reference surface for the declared sector and certificate interfaces. Its release record is v0.1.0, doi:10.5281/zenodo.21219417. This edition’s primitive-standing construction and consumer receipt are mathematical manuscript results. A formalization of this edition must represent the original duty predicate independently of the desired count, prove the joint receiving construction and support minimality, derive the role template and full live-union occupation, and prove the consumer’s actual incidence and reader maps.

A theorem whose endpoint predicate is defined by $n = 3$, or whose proof merely projects a supplied count or rank-one field, does not implement that construction. A source scan for forbidden declarations and a clean axiom report do not by themselves establish manuscript faithfulness. The theorem types, definitions, physical-reception hypotheses, build, and exported dependencies must also be inspected. No Lean build or implementation of the present receiving proof is asserted here.

A manuscript-faithful Lean route for SM-6 represents:

1. the fixed Standard Model scope D_{SM}^3 and the final carrier package C_{SM}^{prim} as explicit carriers;
2. received SM-1 through SM-5 sector packages, the Higgs endpoint, and the generation-cardinality endpoint as typed support records;
3. same-scope, target-update, derivative-skin, projection, transport, route, extension, burden, and failure statuses as explicit predicates;
4. the global coverage theorem and residual-survivor deletion theorem as theorems over declared hypotheses and received packages;
5. BSM, EFT, anomaly, projection, numerical, Higgs, and generation escape routes as finite branch exhaustions rather than informal exclusions;
6. the empty-survivor endpoint $\text{Survprim}_{D_{SM}^3}(\text{Attr}(D_{SM}^3) \setminus C_{SM}^{\text{prim}}) = \emptyset$ as the manuscript-facing closure anchor.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on `axiom`, `sorry`, `admit`, `unsafe`, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

B Imported theorem traceability

This appendix records the dependency anchors imported into SM-6. It is a compact theorem-trace table, not a re-publication of the upstream proofs. The listed anchors identify the imported

statement, its source role, and its use in the SM-6 closure proof. Full verification of an imported theorem belongs to the cited upstream source; SM-6 states exactly what is received and where it is used.

Source role	Imported theorem/status	Statement summary	Trace anchor	SM-6 use
Kernel source	Kernel necessity; non-degenerate construction	Non-degenerate physical claimhood requires preserved admissibility, standing, reference, sameness, and comparison-class conditions. Kernel denial must recover these functions or provide a faithful same-regime counterexample.	kernel necessity; admissibility theorem	kernel status; reality contact
SM-1 master classifier	No primitive skin promotion; target-update separation; BSM audit closure	Same-scope skins, representatives, routes, projections, future-claim surfaces, and BSM audit cases do not become primitive merely by utility or empirical salience; target-changing claims exit the same-scope arena.	classifier closure package	classifier and target discipline
SM-2 gauge sector	Scoped gauge-sector primitive-status deletion	Gauge-sector attributes outside C_{gauge} have no residual primitive standing in the gauge domain.	gauge residual deletion theorem	gauge constructor deletion
SM-3 matter sector	Scoped matter-sector primitive-status deletion	Matter-sector attributes outside C_{matter} have no residual primitive standing under fixed generation governance.	matter residual deletion theorem	matter constructor deletion
SM-4 CCR sector	Scoped charge/coupling/RG primitive-status deletion	Charge, coupling, and RG attributes outside C_{CCR}^{prim} are moduli, transports, calibrations, schemes, projections, or burdens rather than primitive carriers.	CCR residual deletion theorem; CCR status atlas	CCR constructor deletion
SM-5 flavor sector	Scoped flavor primitive-status deletion; generation governance	Flavor/mixing attributes outside $C_{\text{flavor}}^{\text{prim}}$ do not retain primitive transition standing; generation cardinality is governance, not primitive carrier status.	flavor residual deletion theorem; generation-governance theorem	flavor and generation separation
Higgs endpoint	No residual primitive Higgs attribute	Same-scope Higgs-domain attributes outside $H_{br/fix}^{\text{prim}}$ are representatives, routes, comparisons, extension dispositions, burdens, failures, or target updates.	Higgs domain endpoint theorem	Higgs component of C_{SM}^{prim}
Higgs support stack	No fifth identity ground; no primitive Higgs scale; extension/vacuum route classification	Scalar, vev, potential, pole-mass, scale, hierarchy, vacuum-route, extra-scalar, and lower-branch claims do not add independent Higgs primitive standing inside the fixed domain.	Higgs identity, scale, extension, vacuum, and lower-vacuum endpoints	Higgs hardening; BSM/Higgs audit

Structural spine	Structural-spine constructor generation	The fixed Standard Model carrier target is generated by gauge product, matter representation response, charge lattice, Higgs/bridge, Yukawa availability, anomaly compatibility, vacuum compatibility, and parameter-status separation.	standalone SM structural-spine theorem	constructor adequacy and carrier coherence
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C Source inventory

Source	Imported role	SM-6 use
Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model	master classifier; carrier/skin distinction; target-update discipline; burden and hostile-branch grammar	global status discipline and prediction/falsification protocol
Gauge Carriers: Photon, Gluon, and Weak-Boson Standing	gauge-sector primitive closure and C_{gauge}	gauge component of C_{SM}^{prim}
Matter Carriers: Quarks, Leptons, and Neutrinos	matter-sector primitive closure under fixed family governance and C_{matter}	matter component of C_{SM}^{prim} ; route/projection discipline
Charges, Couplings, and Renormalization as Standing Transport	CCR primitive closure and C_{CCR}^{prim} ; registered coupling/RG statuses	charge/coupling component of C_{SM}^{prim} ; parameter discipline
Flavor Transition, Mixing, and Generation Cardinality	flavor/mixing closure and $C_{\text{flavor}}^{\text{prim}}$; generation governance	flavor component of C_{SM}^{prim} ; generation separation
The Higgs Domain Endpoint: Carrier, Identity, Scale, Vacuum, Extension, and Projection	Higgs bridge/fixation endpoint $H_{br/fix}$ and integrated Higgs classifier	Higgs component of C_{SM}^{prim} ; no fifth Higgs primitive
Endpoint on Higgs Structural Identity: Bridge-Carrier Occupation and No Fifth Identity Ground	Higgs identity collapse and no-fifth-ground support	Higgs identity bridge

No Primitive Higgs Scale: Carrier-Indexed Scale Standing and Hierarchy Orientation	no primitive Higgs scale; hierarchy orientation	Higgs scale status discipline
The Higgs Extension Classifier: Extra Scalars and Same-Scope Carrier Preservation	extra scalar and extension classification	BSM/Higgs extension audit
Non-Degenerate Construction and the Kernel of Admissibility	upstream kernel necessity: non-degenerate claimhood, sameness, standing, reference, comparison, prediction, and falsification	kernel-status doctrine; sameness and target-preserving contact
The Standalone Standard Model Structure from First Principles	structural spine: gauge product, representation, Higgs/bridge, Yukawa, charge, anomaly, vacuum, parameter-status structure	carrier architecture coherence and constructor generation

D Dependency ledger

Source	Exact imported theorem or status	Source anchor	SM-6 use	Failure mode
SM-1	Theorem 8.1, No Primitive Skin; Theorem 12.4, Target Update Separation; Theorem 12.6, No Same-Scope Primitive Escape Schema; Theorem 12.7, BSM Audit Closure	Sections 8 and 12	classifier, target-update, BSM/future-claim grammar	classifier break
SM-2	Theorem 21.1, Gauge-Sector Classification; Theorem 21.4, Gauge Carrier Certificate Closure; Theorem 21.5, Scoped Gauge-Sector Primitive-Status Deletion	Section 21	gauge constructor and certificate failure	gauge deletion break
SM-3	Theorem 18.3, Matter Coverage; Theorem 18.9, Matter carrier package closure; Lemma 18.10, Certificate-to-primitive-status deletion; Theorem 18.11, Scoped matter-sector primitive-status deletion	Section 18	matter constructor, mass route, fixed generation envelope	matter deletion break
SM-4	Theorem 17.2, CCR Coverage; Theorem 18.1, Primitive charge-carrier package closure; Theorem 18.2, Registered CCR status package closure; Theorem 18.5, Scoped CCR primitive-status deletion	Sections 17–18	charge/coupling/RG and EFT transport discipline	CCR break

SM-5	Theorem 16.3, Flavor Coverage; Theorem 17.1, Primitive flavor carrier package closure; Theorem 17.2, Fixed generation-governance closure; Theorem 17.6, Scoped flavor-sector primitive-status deletion	Sections 16–17	flavor, PMNS/CKM, CP, generation governance	flavor break
Higgs endpoint	Theorem 8.3, No residual primitive Higgs attribute; Theorem 9.1, Higgs domain endpoint	Sections 8–9	$H_{br/fix}$ and Higgs no-residue import	Higgs endpoint break
Higgs identity support	Theorem 7.3, No fifth identity ground; Theorem 8.1, Higgs Structural Identity Theorem	Sections 7–8	no additional Higgs identity primitive	identity break
Higgs scale support	Theorem 8.2, No Primitive Higgs Scale; Theorem 15.3, Primitive scale deletion; Theorem 15.4, Hierarchy orientation endpoint	Sections 8 and 15	scale/hierarchy nonpromotion	scale break
Higgs extension support	Theorem 15.2, Extension classification theorem; Theorem 16.2, Same-scope extra-Higgs counterexample burden; Corollary 16.3, Primitive extra-Higgs deletion	Sections 15–16	extra scalar and extension audit	extension break
Kernel source	Kernel Necessity / Non-Degenerate Construction theorem; sameness, standing, reference, and comparison-class preservation	Kernel source; DOI 10.5281/zen- odo.20322709	kernel status and reality-contact doctrine	kernel break
Structural spine	Theorem 9.7, Standalone Standard Model Structural Spine; Theorem 12.1, Consolidated Structural Derivation	Sections 9 and 12; DOI 10.5281/zen- odo.20323596	architecture coherence and constructor generation	spine break
SM-6 local	Constructor Adequacy; Carrier-Descent Incompatibility; Role-Necessity Exhaustion; No Valid Same-Scope Hostile Certificate	Main text	global synthesis and final deletion	bridge break

E Carrier register

Carrier	Imported from	Primitive role	Licensed skins/statuses
$C_{gauge-order}$	SM-2	gauge order standing	gauge-field notation, group presentation
$C_{transport}$	SM-2	gauge transport standing	connection displays, propagation routes
$C_{curv/hol}$	SM-2	curvature/holonomy standing	field strength notation, loop displays

$C_{charge-response}$	SM-2/SM-4	response to gauge/charge standing	charge labels, response values
$C_{color-transport}$	SM-2	color transport standing	gluon/color notation
$C_{weak-chiral}$	SM-2	weak-chiral standing	weak labels, chiral bookkeeping
$C_{anomaly-compat}$	SM-2/spine	anomaly compatibility standing	anomaly cancellation displays
$C_{gauge-vacuum}$	SM-2/spine	gauge-vacuum standing	vacuum presentations
$C_{matter-rep}$	SM-3	matter representation standing	quark/lepton symbols
$C_{color-bearing-matter}$	SM-3	colored matter role	quark and hadron labels
$C_{colorless-charged-matter}$	SM-3	charged lepton role	electron/muon/tau labels
$C_{neutral-weak-matter}$	SM-3	neutrino/neutral weak role	oscillation routes, neutrino labels
$C_{mass-response-slot}$	SM-3/Higgs endpoint	mass-response role	mass labels, Yukawa routes
$C_{charge-witness}$	SM-4	witness for charge standing	measurement and calibration
$C_{charge-lattice}$	SM-4/spine	charge lattice standing	charge-number displays
$C_{flavor-transition}$	SM-5	transition standing	CKM/PMNS coordinates
$C_{CP/holonomy-residue}$	SM-5	CP-like residue standing	phase-coordinate skins
$H_{br/fix}$	Higgs endpoint	bridge/fixation carrier	vev, scalar notation, potential, pole mass

F Registered status atlas

Registered status	Examples	Hostile-certificate condition
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Field symbol	A_μ, G_μ, W_μ , spinors, scalar notation	must show symbol carries irreducible role, not representation
Particle label	photon, gluon, weak boson, quark, lepton, neutrino, Higgs	must show label is primitive beyond carrier role
Coupling modulus	g_i, e, α , weak angle	must show numerical carrier role outside C_{CCR}^{prim}
RG transport	beta functions, running curves	must show transport curve is irreducible carrier
Scheme/matching	thresholds, renormalization schemes	must show scheme is primitive, not coordinate choice
Mass/width	quark, lepton, neutrino, Higgs, weak-boson values	must show value itself is primitive carrier
Flavor coordinate	CKM/PMNS entries, mixing angles	must show coordinate is primitive transition ground
Phase coordinate	CP phases, rephasing conventions	must show residue outside $C_{CP/holonomy-residue}$
Higgs representative	vev, potential, scalar field	must show primitive Higgs ground outside $H_{br/fix}$
Vacuum route	metastability, lower branch, tunneling	must show route is primitive standing
EFT/projection	operators, Wilson coefficients, matching surfaces	must show same-scope primitive EFT carrier
Detector/evidence	events, tracks, jets, excesses	must show direct role-occupancy primitive certificate
BSM surface	extra scalar, sterile state, dark sector, GUT model	must show same-scope primitive, not update or extension

G BSM/EFT audit matrix

Claim type	Same-scope default	Default status	Hostile/target condition
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Extra scalar	not automatic	extension surface or burden	hostile if nonderivative $H_{br/fix}$ replacement inside D_{SM}^3 ; update if target changes
Sterile neutrino	not automatic	extension, neutral-sector burden, or target update	hostile only with same-scope primitive matter role
Fourth generation	governed branch	extension/burden/target update	hostile if primitive generation carrier closes
Supersymmetry-like extension	usually target-expanding	extension surface or target update	hostile only if same-scope primitive role remains inside D_{SM}^3
GUT/unification surface	projection or target-expanding	RG/projection/status or target update	hostile if same-scope primitive coupling carrier closes
Dark-sector coupling	usually extension	extension, projection, or update	hostile if same-scope primitive carrier outside C_{SM}^{prim}
EFT operator	yes as display	coefficient, projection, or matching	hostile if operator is primitive carrier, not projection
Composite Higgs	variable	representation, extension, target update, or hostile branch	hostile if same-scope primitive Higgs ground closes
Flavor anomaly	yes as pressure	evidence, burden, or transition	hostile if primitive transition carrier closes
Detector excess	yes as evidence	evidence, anomaly, burden	hostile if direct role certificate closes
Quantum-gravity interface	usually target-updating	target update or projection	hostile only if same-scope primitive inside D_{SM}^3

H Target-update ledger

Claim type	Why it leaves same scope	SM-6 status
New carrier inventory	adds primitive role not governed by D_{SM}^3	target update; hostile same-scope certificate if the fixed target is preserved
New admissibility standard	changes the test for standing	target update
New identity target	replaces the object whose primitive status is being closed	target update
Richer theory domain	extends beyond Standard Model carrier arena	target update
Different comparison class	changes what counts as same-scope primitive	target update
New generation cardinality as domain change	changes G_{gen}^3 rather than submitting within it	target update
Quantum-gravity completion as replacement	changes target physics domain	target update or separate closure problem

I Hostile branch certificate

A hostile certificate must demonstrate:

- (1) the claim is same-scope inside D_{SM}^3 ;
- (2) the claim has primitive role standing;
- (3) the claim is not derivative of C_{SM}^{prim} ;
- (4) the claim is not a registered status in T_{SM}^{reg} ;
- (5) the claim is not governed merely by G_{gen}^3 ;
- (6) the claim is not a target update;
- (7) the role is necessary for same-scope Standard Model standing;
- (8) the role is compatible with the fixed comparison class;
- (9) the certificate is closed rather than a burden, anomaly pressure, suggestive model surface, or projection.

If all nine conditions close, SM-6 is falsified. If any condition fails, the claim is resolved by the registered closure architecture.

J Constructor partition ledger

Constructor	Examples	Closure source	Why hostile validity fails
$\mathcal{A}_G$	gauge-field notation, curvature, holonomy, gauge labels	SM-2 plus sector embedding	primitive/non-descent fails
$\mathcal{A}_M$	quarks, leptons, neutrinos, tracks, jets	SM-3 plus generation governance	labels/routes are nonprimitive
$\mathcal{A}_{CCR}$	charges, couplings, RG curves, schemes	SM-4	registry/non-descent fails
$\mathcal{A}_F$	CKM, PMNS, phases, oscillations	SM-5	coordinate/status fails
$\mathcal{A}_H$	scalar notation, vev, potential, vacuum branch	Higgs endpoint	all non- $H_{br/fix}$ claims descend or register
$\mathcal{A}_{gen}$	family labels, cardinality branches	SM-3/SM-5 governance	nongovernance fails
$\mathcal{A}_{num}$	masses, widths, values, fits	numerical nonpromotion	enters registered status
$\mathcal{A}_{meas}$	events, excesses, detector records	measurement nonpromotion	evidence is not primitive role
$\mathcal{A}_{proj/EFT}$	Wilson coefficients, matching, projections	projection/EFT role-failure	role necessity or registry fails
$\mathcal{A}_{BSM/anom}$	new sectors, anomalies, model surfaces	BSM certificate-failure	routes through constructors
$\mathcal{A}_{int}$	cross-sector interface, matching, translations	interface role-failure	shared carrier, registry, burden, or failed role
$\mathcal{A}_{ov}$	mixed numerical/ flavor/Higgs/EFT claims	overlap no-promotion	multiple dependencies do not sum to primitiveness

K Theorem inventory

No.	Name	Dependency	Role in final proof
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1	Attribute Membership Lock	SM-6 formal arena	equates $\text{Attr}(D_{SM}^3)$ with same-scope SM claims
2	Constructor Exhaustion	constructor presentation	makes case partition formal
3	Constructor Adequacy	carrier architecture	prevents arbitrary closure by definition
4	Shared-role coherence	SM-2, SM-4	prevents duplicated charge-response primitives
5	Union does not generate primitives	SM-6 local	prevents integration primitive smuggling
6	Sufficient carrier package	all sector imports	sets C_{SM}^{prim} for theorem
7	Global status registration	SM-1 through SM-5, Higgs	registers nonprimitive statuses
8	No registered-status promotion	SM-1, SM-6 local	blocks skin promotion
9	Generation governance separation	SM-3, SM-5, open-generation endpoint	keeps G_{gen}^3 outside C_{SM}^{prim} while instantiating it at three
10	Fourth-generation branch classification	SM-3, SM-5	closes generation escape route
11	Carrier architecture coherence	structural spine	shows package is one architecture
12	Sector residual deletion preservation	all sector imports	imports local closure globally
13	Cross-sector certificate exposure	SM-6 local	preserves falsifiability
14	Projection-interface nonpromotion	SM-1, SM-6 local	blocks mediation promotion
15	Projection role-failure	SM-6 local	identifies failed validity conjuncts for projections
16	EFT display classification	SM-4, SM-6 local	classifies EFT objects
17	EFT role-failure	SM-6 local	blocks EFT certificate validity
18	Wilson coefficient status	SM-4, SM-6 local	keeps coefficients registered

19	Interface role-failure	structural spine, SM-6 local	blocks interface primitive multiplication
20	Numerical availability nonpromotion	SM-4, SM-5, Higgs	blocks parameter objection
21	Measurement nonpromotion	SM-6 local	blocks evidence-to-carrier promotion
22	No fifth Higgs primitive	Higgs endpoint	closes Higgs residual escape
23	Higgs extension classification	Higgs extension support	classifies extra scalar claims
24	BSM target-update boundary	SM-1	separates target-changing BSM claims
25	Same-scope BSM exhaustion	SM-1, all sectors	audits same-scope BSM surfaces
26	Same-scope BSM certificate-failure	constructors plus BSM theorem	blocks BSM as magic escape class
27	Anomaly nonpromotion	SM-1, SM-2, SM-6 local	blocks anomaly-report promotion
28	Target-update boundary	SM-1	separates new target domains
29	Hostile certificate nonabsorption	SM-1, SM-6 local	exposes falsification
30	Global same-scope coverage	constructors plus all prior results	assigns every same-scope attribute
31	Constructor partition theorem	Attribute Lock and Constructor Exhaustion	prevents unlisted same-scope classes
32	Overlap no-promotion	SM-6 local	blocks multi-status leakage
33	No registered-to-primitive leakage	status atlas	converts coverage to deletion
34	Governance no-leakage	generation governance	blocks generation leakage
35	Burden and failure no-leakage	hostile protocol	blocks burden/failure promotion

36	Target-update no-leakage	target boundary	removes target updates from survivors
37	Role-Necessity Exhaustion	constructor adequacy, sector closures	necessary roles are in or descend to C_{SM}^{prim}
38	Sector constructor certificate failure	sector closures and Higgs endpoint	blocks sector hostile validity
39	Registered and governed constructor certificate failure	status/governance lemmas	blocks value, evidence and governance validity
40	Projection/EFT constructor certificate failure	projection/EFT role-failure	blocks hard projection/EFT cases
41	BSM/anomaly constructor certificate failure	BSM certificate-failure	blocks BSM/anomaly hard case
42	Interface and overlap constructor certificate failure	interface/overlap lemmas	blocks mediation/overlap cases
43	No Valid Same-Scope Hostile Certificate	certificate-conjunct failures	eliminates hostile validity inside D_{SM}^3
44	Hostile branch closure inside fixed arena	no-valid-certificate theorem	closes branch attempts without absorption
45	Standard Model Carrier Exhaustion Closure	all prior results	final residual deletion theorem

L Decision-tree worksheet

Step	Question	Yes	No
1	Is the claim Standard Model-associated?	Continue	External to SM-6 claim arena
2	Is it same-scope in D_{SM}^3 ?	Continue	Target update
3	Is it in C_{SM}^{prim} ?	Carrier member	Continue
4	Does it descend to C_{SM}^{prim} ?	Derivative carrier descent	Continue
5	Is it registered in T_{SM}^{reg} ?	Registered nonprimitive	Continue
6	Is it generation-governed?	Governance nonprimitive	Continue
7	Does it alter the target?	Target update	Continue

8	Does it claim primitive status without certificate?	Counterexample burden	Continue
9	Does it fail its role?	Failed primitive	Continue
10	Does it close a same-scope primitive certificate outside C_{SM}^{prim} ?	Validity is ruled out inside D_{SM}^3 ; if supplied externally, it defeats the theorem	No primitive survivor

M Primitive-survivor audit worksheet

Certificate requirement	Audit question	Required
Same-scope	Does the claim remain inside D_{SM}^3 without changing domain or target?	yes
Outside carrier package	Is the claim outside C_{SM}^{prim} ?	yes
Primitive role	Does the claim occupy an irreducible carrier role?	yes
Nonderivative	Is it not licensed by, measured through, or projected from C_{SM}^{prim} ?	yes
Nonregistered	Is it not a value, coordinate, route, projection, representative, measurement, anomaly, or model surface?	yes
Nongovernance	Is it not merely generation-governed?	yes
Not target update	Does it avoid changing comparison class or admissibility?	yes
Role necessity	Is the role necessary for same-scope standing?	yes
Closure	Is the certificate closed rather than burden-bearing?	yes

N Evidence-tier register

Tier	Name	Examples	Primitive effect
E1	refinement	improved mass, coupling, width, branching ratio, fit	none; registered status
E2	anomaly pressure	detector excess, flavor tension, cosmological tension	burden or repair demand
E3	extension surface	BSM model explaining an anomaly	extension, burden, or projection

E4	target update	new regime or changed carrier domain	outside D_{SM}^3
E5	hostile certificate	same-scope primitive role outside C_{SM}^{prim}	falsifies theorem if valid

O Detailed nonclaim ledger

Nonclaim	Reason it is not needed for SM-6
Empirical finality	Primitive carrier closure is not finality of nature.
No new physics	New physics may be target update, extension, projection, anomaly, or hostile certificate.
Numerical derivation of all masses	Mass values are registered response/fixation/measurement statuses.
Numerical derivation of all couplings	Couplings are moduli, calibration values, and transport statuses.
Derivation of CKM/PMNS entries	Entries are transition coordinates or boundary reconstructions.
Derivation of CP phase values	Raw phase values are coordinates; residue status is separately carried.
Empirical derivation of generation count	Three-generation governance is received from primitive necessity and complete occupation through Proposition 3.1; family labels and matrix dimension are not count selectors.
Model-by-model BSM exclusion	SM-6 classifies BSM status rather than cataloguing all models.
Elimination of EFT language	EFT remains registered projection/matching/coefficient language.
Closure of all future anomaly outcomes	Future anomalies are evidence pressure, burden, target update, or hostile certificate.

P Closure audit checklist

Audit item	Resolution	Status
Strong endpoint retained	final theorem is residual primitive deletion, not classifier-only discipline	closed

Empirical finality avoided	nonclaims and target-update openness stated repeatedly	closed
G_{gen}^3 kept outside C_{SM}^{prim}	generation governance section and theorem	closed
T_{SM}^{reg} kept outside C_{SM}^{prim}	registered-status promotion theorem	closed
Higgs endpoint imported	$H_{br/fix}$ included and no-fifth-Higgs theorem stated	closed
Projection/interface gap closed	projection, EFT, and interface role-failure lemmas	closed
BSM audit included	BSM exhaustion, certificate-failure lemma, and matrix	closed
Hostile branch exposed	nonabsorption, no-valid-certificate theorem, and template	closed
Role-necessity exhaustion	necessary same-scope roles are in or descend to C_{SM}^{prim}	closed
Certificate-conjunct table	each constructor blocks at least one validity conjunct	closed
No numerical overclaim	parameter-status theorem and nonclaims	closed
